AION-OMEGA^Q / QSIGMA

HypercomplexBrain

A human guide to the architecture, evidence, experiments, and honest frontier



Worldlets share a base, keep local geometry, and expose remainder.

Repository by Jesús José Vilela Jato

Publisher-review edition v2.0 · branch claude/aion-brain-architecture-27f8ji · commit fbf7b9b · 28 August 2026

Open the inspected branch

Preface to the publisher-review edition

A reader-first companion that preserves the repository's correction history and evidence boundaries.

HypercomplexBrain contains a large vocabulary of custom objects, mathematical analogies, executable instruments, experiments, corrections, and deliberately unpromoted hypotheses. Its difficulty is not only technical. The reader must also learn which words name implemented software structures, which name mathematical surrogates, which refer to measurements, and which remain design language.

This edition therefore follows one repeated explanatory sequence: intuition, formal object, implementation surface, evidence status, limitation, and failure condition. The goal is not to make the programme sound simpler than it is. The goal is to make its complexity inspectable by a reader who did not participate in its construction.

EDITORIAL PURPOSE

The central editorial rule is that fluency must not erase epistemic friction. Whenever an idea becomes easier to read, the place where it can fail should become easier to see as well.

How to read this guide

You do not need prior familiarity with this research programme. A mathematically curious reader can begin with Chapters 1, 2, 7, 8, 17, and 22. A software reader can follow the claim-to-artifact path in Chapters 3, 6, 20, and 21. A specialist can move directly to the geometry, quantum, and experimental chapters, while keeping the P/A/M/H/S/R evidence tags in view.

Reader	Recommended path	What the path provides
New to the programme	Executive summary -> 1 -> 2 -> 7 -> 8 -> 22	A concrete model of the architecture, its custom vocabulary, and its strongest boundaries.
Software engineer	3 -> 6 -> 7-11 -> 20 -> 21	The route from types and transports to tests, manifests, artifacts, and correction gates.
Mathematician / physicist	6 -> 9-16 -> 19 -> 20	The geometry, coherence surrogates, hypercomplex operators, quantum surfaces, and theorem gaps.
Research collaborator	2 -> 17-23 + command appendix	The active frontier, negative results, and the disciplined way to propose a new round.

Editorial and evidence note

SNAPSHOT BOUNDARY

This guide is grounded in committed branch material through fbf7b9ba8974364ee212a57e347ab0d0de37e454. It incorporates the R1Z preregistration and finding, but it does not claim independent reproduction, cross-platform verification, or a fresh rerun of the full suite. Repository statements about local test counts remain repository reports unless separately reproduced.

Field	Value
Repository	jesusvilela/hypercomplexbrain
Inspected branch	claude/aion-brain-architecture-27f8ji
Snapshot commit	fbf7b9ba8974364ee212a57e347ab0d0de37e454
Package surface	aionq 0.1.0; Python >= 3.10; NumPy/SciPy/Qiskit; pytest development suite
Current frontier	QSigma-R1Z executed locally; growth measurements recorded; criterion defect preserved; asymptotic claim remains H
Review status	Developmental editorial review plus peer-style technical audit; not anonymous independent peer review
Repository mutation	None. This edition is a local document only.

About the author

Jesús José Vilela Jato is a senior engineering leader and independent researcher working across artificial intelligence, data platforms, geometric deep learning, and evidence-governed computational research. HypercomplexBrain is presented here as an evolving research programme rather than as a claim of completed general intelligence or a closed mathematical theory.

Abbreviations and symbols

Term	Meaning in this guide
AION-ΩQ / QΣ	The umbrella architecture and its evidence-governed quantum/geometry research sequence.
HB-0	The executable polycosmic kernel and its reference factory.
P / A / M / H / S / R	Proved, Axiomatized, Measured, Hypothesis, Semantic, Retired.
W_i	A Worldlet: one typed local context.
T_ij	A Tendril: a declared translation/correspondence from Worldlet i to j.
[R]_x	A typed remainder: an unresolved object whose channel and evidence status remain attached.
B = H^n	The shared Poincaré base used as an A-tagged architectural coordinate contract.

Term	Meaning in this guide
δ	A measured defect; its type and instrument must be named before comparison.

AI-assisted editorial preparation

This local edition used AI-assisted editorial analysis, repository inspection, document production, and layout checking. The author remains responsible for the scholarly claims, source selection, permissions, interpretation, and any future submission. The repository artifacts, not this explanatory layer, remain the authority for executable statements.

Contents at a glance

Part	Theme	Chapters
Part I	Orientation	1. The repository in one sentence; 2. The evidence grammar; 3. Repository anatomy
Part II	Why the architecture looks this way	4. The failure of scalar quantumness; 5. The Atlas; 6. Frames, transport, and remainder
Part III	Worldlets and local-world architecture	7. Worldlets; 8. Tendrils; 9. Cages and coherence; 10. Identity and recognition; 11. Quality, governance, and time
Part IV	Geometry, quantum, and information surfaces	12. Shared H^4 base; 13. N-ary decision field; 14. Quantum fibers; 15. Memory and visibility; 16. Fano and finite topology
Part V	What the experiments actually say	17. Correction timeline; 18. R1W competence; 19. R1X-R1Z frontier
Part VI	Reproducibility and use	20. Formalization and witnesses; 21. Run and extend
Part VII	Assessment	22. Surviving claims; 23. Smallest honest road forward

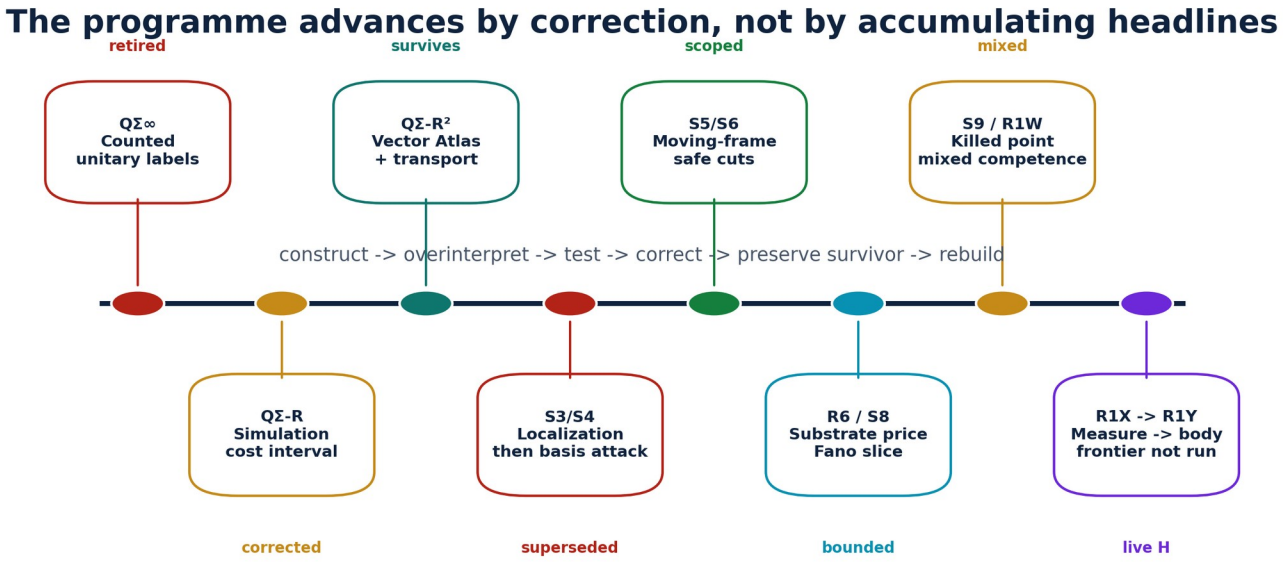
Executive summary

THE SHORTEST ACCURATE DESCRIPTION

HypercomplexBrain is an evidence-governed research harness for testing quantum, geometric, hypercomplex, and agentic architecture ideas. Its strongest demonstrated capability is not consciousness or quantum advantage. It is the ability to turn ambitious language into typed operators, run controls against those operators, and preserve corrections when the controls win.

1. The repository is not an implemented biological brain, a conscious system, or an AGI claim. “Brain,” “identity,” “memory,” and “recognition” name engineering structures and measurable operators.
2. The research object has three layers: an architecture vocabulary, executable mathematical and numerical instruments, and a governance system that controls what can be published as proved, axiomatized, measured, hypothesized, semantic, or retired.
3. The project abandoned scalar “quantumness” because a single number repeatedly gave persuasive but mistyped answers. The replacement is a vector-valued Quantum Necessity Atlas whose coordinates retain uncertainty, free-theory dependence, frame, and missingness.

4. HB-0 models cognition as a network of Worldlets rather than one global vector. Worldlets may use different local geometries and carriers, but they share an A-tagged Poincare H^4 base and must translate through explicit, reversible-or-loss-reporting Tendrils.
5. The architecture refuses flattening in several places: sixteen mind-quality channels cannot be reduced by a weighted sum; energy is a vector budget; time has fast, mesoscopic, and slow clocks; n-ary decisions retain higher cells and all Fin(n) sectors rather than being assembled from pairwise winners.
6. The quantum components are carefully scoped. Several exact or numerical identities survive, including the unitary price of noncompact depth and a compact chiral SU(2)+ spin transport over H^4. These are simulator and representation results, not hardware or speedup claims.
7. The experimental record contains meaningful negative results. S4-G100 failed six of ten gates; S9 repaired its numerical frame but still killed the frozen parameter point; R1W passed tracking and coordination components but failed its combined promotion gate.
8. The correction ledger is load-bearing. Earlier results remain visible with supersession links, so a later reader can reconstruct not only what is believed but why earlier interpretations were abandoned.
9. The formal surface is mixed rather than globally verified. One narrow Lean file has a direct local acceptance record; another is dependency-blocked and contains a sorry; the vendored corpus is not a completed proof surface.
10. At the inspected snapshot, the live frontier is R1Z. The round correctly moved from a fixed-register spectral picture to a size-dependent growth question in the category of fermionic quasi-free states. Its most important result is methodological: the preregistered BOUNDED verdict was unreachable for genuinely bounded-looking sequences, and the frozen zero-count observable required a degeneracy condition that 32 of 105 rows did not satisfy. The failed predictions remain failed; the rule is not patched post hoc; no asymptotic dichotomy is claimed.



A compressed view of the programme: attractive constructions are repeatedly reframed by controls, then preserved rather than erased.

Three lenses used throughout

Lens	Question	Reader discipline
Architecture	What structure is being proposed?	Preserve the intuition, but do not inherit evidence from the vocabulary.
Instrument	What operator or measurement makes the proposal falsifiable?	Inspect its type, frame, assumptions, controls, and numerical domain.
Governance	What may the result authorize?	Follow the evidence tag, active remainder, witness state, and supersession history.

READING PROMISE

Every major architectural idea in this guide is paired with its executable meaning and its evidence ceiling. The purpose is understanding, not promotional compression.

PART I

Orientation

What the repository is, how it speaks, and where its authority comes from

observe | verify | leave the remainder visible

CHAPTER 01

The repository in one sentence

A research laboratory whose primary product is a disciplined relationship between imagination and evidence.

A newcomer can be forgiven for reading the name “HypercomplexBrain” and expecting a neural network, a cognitive agent, or a speculative manifesto. The repository is none of those in isolation. It is closer to a compact research institute encoded as software: a place where architectural ideas are translated into mathematical objects, those objects are made executable, and every attempted promotion is forced through explicit controls.

The repo has an unusual dual character. One side is generative: it uses language such as Worldlet, Tendril, polyholonomic identity, recognition, resonance, holoportation, Bordón respiration, and polycosmic kernel. These terms are not decoration. Each is intended to preserve a structural intuition that conventional flat pipeline language would prematurely erase. The other side is adversarial: the code insists on types, transports, nulls, resource parity, frame attacks, preregistration, kill conditions, and corrections. The creative language proposes; the evidence machinery disposes.

This makes the repository easier to misunderstand from both directions. A purely lyrical reading will overclaim. A purely conventional software reading will miss why the abstractions were introduced. The right reading is braided: semantic architecture, mathematical operator, implemented instrument, and evidence label must be followed together.

HUMAN TRANSLATION

Think of the repository as a wind tunnel for unusual architectures. A Worldlet, hyperbolic base, quaternion relation, or quantum fiber is a proposed airframe. The tests and controls do not ask whether the airframe looks advanced; they ask where it stalls, what is invariant, and whether a simpler shape flies just as well.

The three braided layers

Layer	What lives there	How to read it
Architecture	Worldlets, Tendrils, identity ecology, quality fields, higher cells, quantum fibers	As a structured design hypothesis. It becomes evidence only through a typed operator and experiment.
Instrumentation	Atlas coordinates, transport defects, rank losses, Chern probes, simulation-cost bounds, visibility depth	As executable definitions with assumptions, numerical domain, and controls.
Governance	P/A/M/H/S/R, manifests, envelopes, append-only artifacts, witness weave, supersession	As the authority system that decides what may be said publicly.

What the name “brain” does and does not mean

In this codebase, “brain” names an architectural ambition: multiple local worlds, partial viewpoints, memories, translations, resource limits, self-models, and coordination surfaces coexisting without being forced into one state vector or one global clock. That is a legitimate software and mathematical research programme.

It does not establish subjective experience, sentience, consciousness, human-like understanding, biological fidelity, or general intelligence. The repository states this boundary explicitly, and the guide preserves it. The strongest live reading is “a typed architecture for studying conditions under which richer coordination may help,” not “an artificial mind.”

THE STRONGEST PRESENT ACHIEVEMENT

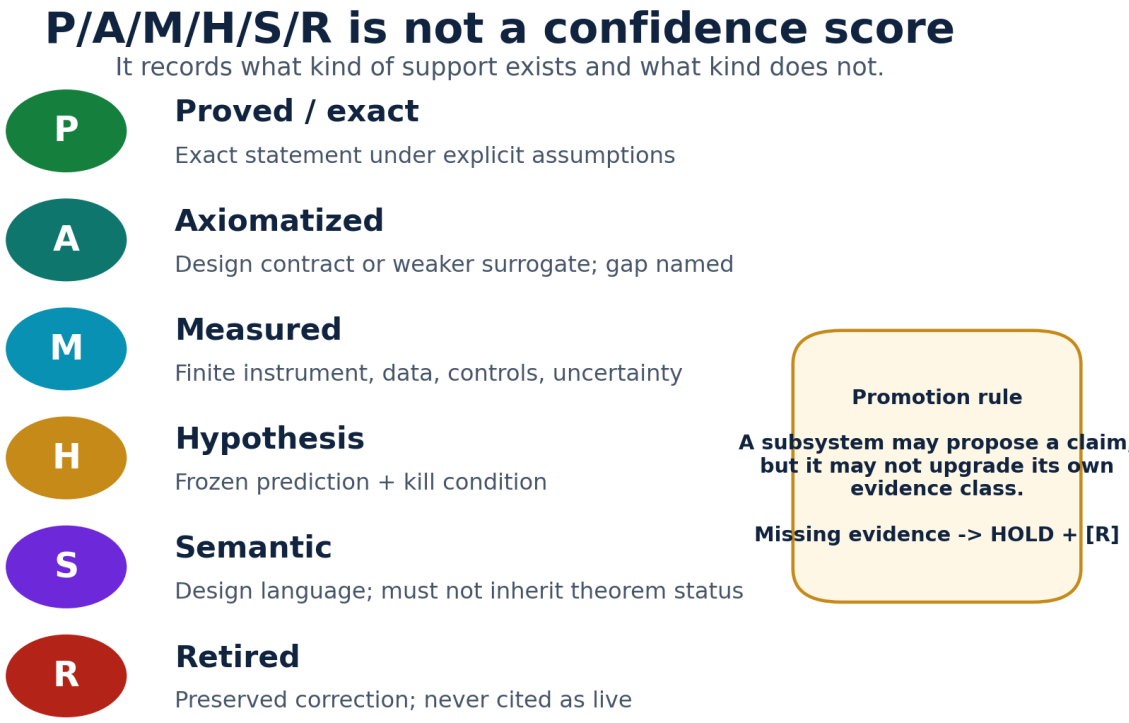
The repository has made its own tendency toward overinterpretation observable. It records when a beautiful object is only semantic, when a theorem is conditional on an axiom, when a measurement is finite and self-attested, and when a prior claim must be retired. That reflexive discipline is itself a substantive engineering result.

Repository source trail: README.md; src/aionq/__init__.py; docs/ARCHITECTURE.md; docs/HB0_POLYCOSMIC_KERNEL.md

CHAPTER 02

The evidence grammar

P/A/M/H/S/R is a type system for claims, not a confidence dial.



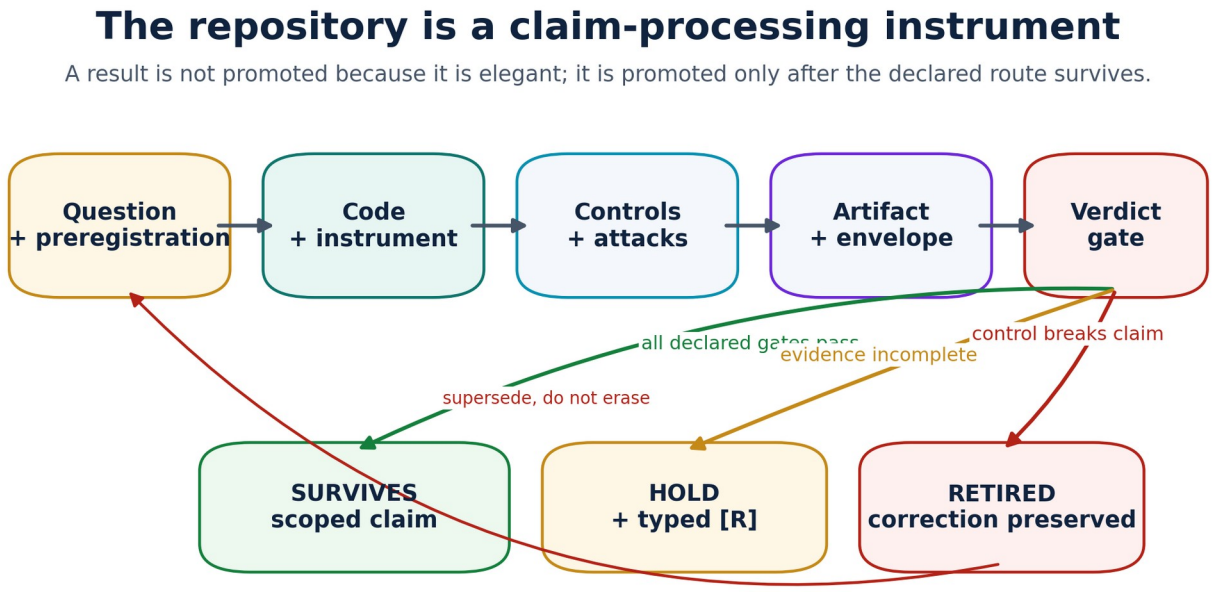
The evidence alphabet records the kind of support that exists. It is deliberately not a ranking of how impressive a claim sounds.

Most research repositories distinguish “working” from “not working.” HypercomplexBrain needs a finer language because its artifacts cross mathematics, numerical experiments, architectural semantics, and speculative bridges. A formula can be exactly correct inside a declared model while the proposed interpretation remains unproved. A numerical result can be reproducible over a finite grid while saying nothing about asymptotic behavior. A semantic object can be useful without being true in any empirical sense.

The repository therefore assigns every claim one of six evidence types. P means proved or exactly computed under explicit assumptions. A means axiomatized, or verified only through a weaker surrogate. M means measured with a bounded instrument and scope. H is a falsifiable hypothesis or open bridge. S is semantic or architectural design language. R is retired by correction, scaling, bug analysis, or matched control.

The A tier is especially important. It prevents a common category error: a compiled program or formal theorem may establish the statement that was encoded, while the intended external meaning depends on an unproved bridge. A makes that bridge visible rather than allowing the surrounding prose to smuggle it in.

Promotion is a transaction



A claim moves through a frozen question, implementation, controls, artifact and verdict. A failed control creates a correction path, not an erased history.

A governed round begins by freezing the question, variables, thresholds, control family, budget, and kill condition. The code then implements both the target instrument and the matched controls. The result is bound to an artifact and an evidence envelope. Only after those pieces exist does the verdict gate evaluate whether the claim survives, remains on hold, or is retired.

The repository also enforces an anti-self-promotion principle. A subsystem may produce evidence, but it cannot simply declare that evidence stronger than it is. Independent review and independent recomputation are different acts; a persuasive review can identify a defect, but it does not become an independent numerical witness. A circular ring of attestations cannot manufacture evidence rank.

Typed remainder: the honest output of incompleteness

A missing component is not represented as zero, NaN hidden in a score, or a vague caveat. It is carried as a typed remainder [R]. A transport remainder, empirical residual, identity remainder, finite-size remainder, witness remainder, and semantic remainder are different objects. The system prohibits silently transporting one kind into another.

This matters because many apparent breakthroughs are created by collapsing “unknown” into “good enough.” Here, missing evidence usually produces HOLD plus an explicit remainder. That is not failure to decide. It is a decision that the current evidence does not authorize a stronger statement.

A PRACTICAL READING RULE

Whenever a sentence feels exciting, ask four questions: What is the exact object? What is its evidence tag? What control could kill it? Which remainder is still active? The repository is designed so those questions have inspectable answers.

Repository source trail: README.md (Evidence); docs/SOURCES_DIGEST.md; src/aionq/governance.py; docs/WITNESS_WEAVE.md

CHAPTER 03

Repository anatomy

Documents specify; code implements; experiments risk claims; artifacts preserve what happened.

The easiest way to navigate the repository is to follow authority rather than file size. README.md is the entrance. It summarizes the current posture and points to the deeper contracts. The docs directory contains derivations, result narratives, correction histories, and interpretation ceilings. The manifests directory freezes questions before execution. The src/aionq package contains the operators. Experiments execute the frozen questions. Tests attack both implementation and publication rules. Artifacts record outputs, while evidence envelopes bind outputs to code, inputs, environment, and witness state.

No single directory is sufficient. Reading only src/aionq shows what the software can calculate but not what the result is allowed to mean. Reading only docs can miss whether a sentence is backed by an executable gate. Reading only tests can confuse a regression contract with a discovery. The intended unit of understanding is a claim graph: claim -> manifest -> implementation -> experiment -> artifact -> envelope -> correction history.

Directory / surface	Role	Trust question
README.md	Current entrance and high-level status	Is the statement scoped and linked to its receipt?
docs/	Architecture, derivations, limits, result narratives	Is this current, historical, or superseded?
manifests/	Preregistration, frozen parameters, kill conditions	Was it committed before implementation or execution?
src/aionq/	Executable operators and typed data structures	What assumptions and numerical domain are encoded?
experiments/	Runners for bounded questions and attacks	Does the command match the frozen manifest?
tests/	Algebraic, numerical, governance, and regression gates	Is the test a control, a theorem surrogate, or a result gate?
artifacts/	Append-only or versioned result payloads	What exactly ran, with which seed, environment, and digest?
formal/	Lean surfaces and provenance notes	Which file actually compiles, with what dependencies and admissions?

A claim’s life cycle

1. A semantic idea is named without pretending it is already mathematics.
2. The idea is translated into objects, operators, invariants, and failure modes.

3. A manifest freezes the discriminating question before the result is available.
4. Code implements target and controls; tests assert invariants and refusal conditions.
5. An experiment emits raw data and a verdict ledger; no result is inferred from presentation alone.
6. An envelope binds provenance and witness state; validity does not imply independent reproduction.
7. Later work may supersede the interpretation while preserving the original artifact and back-pointer.

WHY APPEND-ONLY BEHAVIOR MATTERS

Some experiment producers intentionally refuse to overwrite an existing result. That refusal is not inconvenience: it prevents a rerun with subtly changed code or parameters from silently replacing the evidence that originally supported a claim.

Two useful reading paths

Concept-first path

```
README.md
  -> docs/ARCHITECTURE.md
  -> docs/SOURCES_DIGEST.md
  -> one round document
  -> its manifest, experiment, artifact, tests
```

Audit-first path

```
manifests/<round>.json
  -> src/aionq/<operator>.py
  -> experiments/run_<round>.py
  -> artifacts/<round>/...
  -> tests/test_<round>.py
```

Repository source trail: README.md (Explore and Run); pyproject.toml; repository tree; docs/ARCHITECTURE.md

PART II

Why the architecture looks this way

The design is a record of what simpler instruments failed to preserve

observe | verify | leave the remainder visible

CHAPTER 04

The failure of scalar quantumness

The project became more credible when its headline number stopped working.

The early architecture tried to answer a seductive question with a scalar: how quantum is this design? The first metric counted the weight of edges written as unitary operations. On all-unitary cores it returned 1.0 by construction. Classical reversible circuits, GHZ states with zero magic, and random Clifford circuits could all look maximally quantum under that definition.

The failure was not poor calibration. The metric was constant over the very population it was meant to rank. Local notation had been mistaken for global computational necessity. A collection of quantum-looking transitions can still admit an efficient classical description.

The next instrument priced known classical simulation strategies. That was better, but it inherited a deeper defect: any portfolio of simulators is only as complete as the strategies included. Product chains containing many T gates could look expensive until tensor-network structure was added. T-saturated matchgate circuits could look hard to both stabilizer and contraction estimators while remaining free-fermion simulable. One SWAP control could change that status. Each added strategy improved the bound while simultaneously proving the previous bound was not an intrinsic measure of quantumness.

HUMAN TRANSLATION

Counting “quantum” instructions is like judging a language by the number of exotic words in its source code. Pricing one compiler is better, but the result still depends on which compilers you know. The repository eventually stopped pretending that one scalar could carry all of these distinctions.

The moving-frame attack

A further correction came from relabelling wires. The physical circuit remained the same, but a free-fermion validity test depended on whether interacting wires were adjacent in the chosen numbering. Under a swap of two names, the reported simulation bound could move dramatically. The calculation was not necessarily numerically wrong; it was typed incorrectly as an invariant of the circuit when it was partly an invariant of the representation.

This distinction - wrong value versus wrong type - is central to the repository. A frame-resolving quantity may still be useful, but it must be reported as a coordinate-dependent diagnostic and transported before comparison. It cannot be promoted as if it described the underlying object independent of representation.

Instrument	What it saw	Why it failed as a sovereign score	What survived
Q_naive	Unitary-labelled edges	Constant 1.0 on all-unitary cores; notation confused with necessity	A historical counterexample and tests preventing recurrence
q_sim_upper	Best cost among implemented classical strategies	Portfolio incomplete; some strategies frame-dependent	A typed upper bound relative to declared free theories
Single weighted promotion score	Combined desirable outcomes	Could hide a failed dimension behind a strong unrelated component	Vector outcome plus componentwise promotion cone

DESIGN CONSEQUENCE

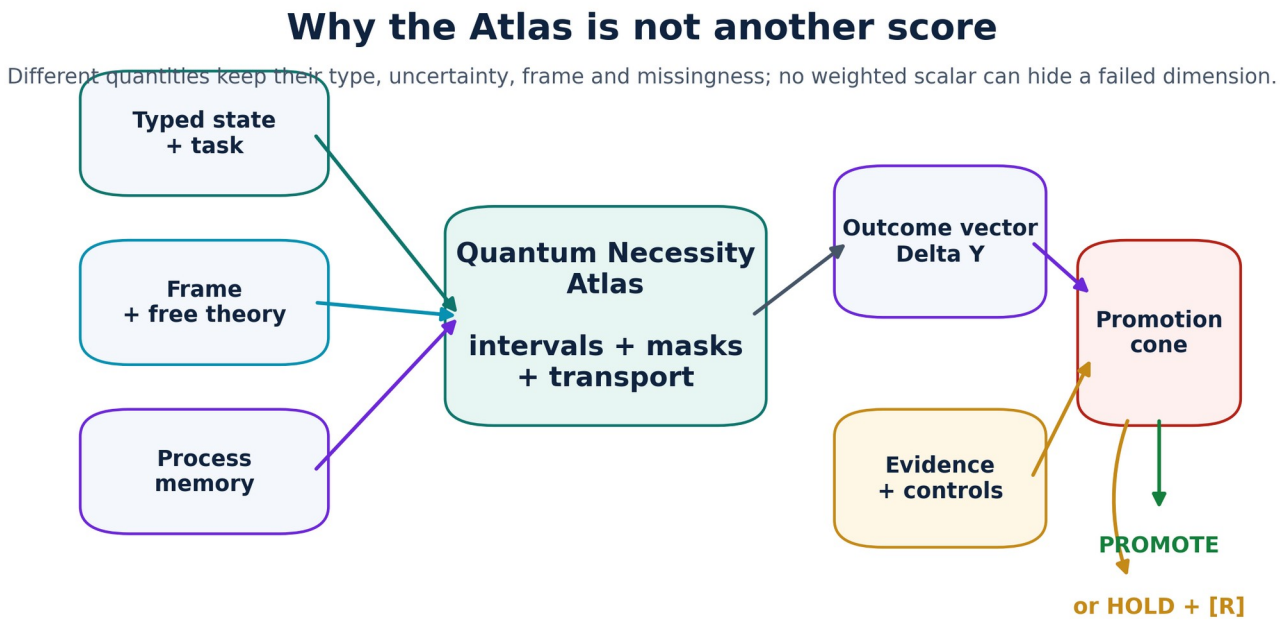
The architecture no longer asks for “the” quantumness of a core. It asks a family of typed questions: relative to which free theory, under which frame, within which interval, with which missing coordinates, and at what simulation or representation cost?

Repository source trail: docs/ARCHITECTURE.md sections 0-0.35; src/aionq/residency.py; src/aionq/contraction.py; src/aionq/movingframe.py

CHAPTER 05

The Quantum Necessity Atlas

A vector-valued map replaces the monarchy of one number.



The Atlas preserves multiple inputs and multiple outcome coordinates. Promotion is a region over the vector, not a weighted average.

The Quantum Necessity Atlas is the architectural answer to the scalar failures. It does not merely add more metrics. It changes the type of the question. A result is a section over declared free theories, with intervals, observation masks, transport rules, and evidence obligations. Two values cannot be compared just because both happen to be floats.

A free theory names what is considered cheap or classically available for the comparison. Stabilizer, tensor-network, free-fermion, basis-permutation, or other simulable structure may each define a different explanatory frame. Quantumness is therefore relative to what the comparison permits for free. The Atlas makes that relativity explicit rather than hiding it in an implementation detail.

The output is a vector Delta Y. Each coordinate can carry its own unit, uncertainty, direction of preference, and missingness. A promotion cone specifies the region in which every required outcome is acceptable. This is stricter than a weighted sum: excellence in one coordinate cannot cancel an identity violation, missing evidence, uncontrolled resource increase, or failed safety gate.

Four invariants that organize the Atlas

Invariant	Plain meaning	Failure prevented
Type before arithmetic	Do not compare bits, strengths, costs, ranks, and residuals because they share a numeric encoding.	False equivalence and accidental unit conversion
Transport before comparison	Values measured over different frames or free theories require an explicit map that names what is preserved or lost.	Frame artifacts presented as object invariants
Whole eigenspaces before arbitrary cuts	At degenerate boundaries, move to a real spectral gap rather than slicing an eigenspace by index.	Unstable basis-dependent measurements
Cone before score	All declared dimensions must clear their gates; one large gain cannot hide one failed obligation.	Goodharting through weighted aggregation

Intervals and missingness are first-class

Many architecture comparisons are bounds rather than exact values. A simulation-cost coordinate may be an upper bound supplied by the best strategy currently implemented. A stronger future classical algorithm may lower it. The Atlas therefore distinguishes what is known from what is merely not yet simulated cheaply.

Likewise, an unmeasured coordinate is not zero. Observation masks and typed remainders carry the difference between “no effect,” “not observed,” and “not meaningful under this free theory.” This discipline is particularly important in the four-cell and witness surfaces, where missing outcomes remain explicit blockers rather than disappearing into a partial average.

WHAT THE ATLAS DOES NOT PROVE

A well-designed measurement surface is not itself evidence of advantage. The Atlas is ready to receive evidence; it does not manufacture cognitive, quantum, or engineering benefit by existing.

Repository source trail: README.md (Map); src/aionq/atlas.py; src/aionq/advantage.py; src/aionq/quantities.py; src/aionq/stack.py

CHAPTER 06

Frames, transport, and typed remainder

The same object can look different from different coordinates; the repo insists on paying for the change of view.

A frame is the coordinate system, labeling, basis, gauge, or local chart through which an object is observed. In ordinary code, frames are often treated as formatting. In this repository they can change measured quantities, numerical stability, and even which classical strategy applies. That makes frame handling a scientific obligation.

The moving-frame machinery distinguishes orbit-collapsed measurements from orbit-resolving ones. An orbit-collapsed quantity remains constant under the declared group action and can be interpreted as a property of the object at that level. An orbit-resolving quantity changes with the frame. It may still be informative, but it is a coordinate reading and must be typed as such.

Transport is the explicit operation that allows comparison across frames, local worldlets, or free theories. A transport records what is preserved, transformed, lost, and paid. Its round-trip defect measures whether returning through the inverse path reconstructs the original state. A convenience fidelity such as exp(-defect) can summarize the number but may not replace the raw defect or erase its type.

State transport is not summary transport

One of the architecture's defining corrections came from a loop across free theories. The underlying state could close exactly, while a lossy summary section failed catastrophically. Earlier prose treated transport as one object. The correction separated state transport from section transport.

The conceptual lesson is broad: a reversible world may have an irreversible description. If an agent compresses a local state into a summary and moves only the summary, information can disappear even when the underlying dynamics is invertible. The repository therefore treats retained state, section, and correction history as different carriers.

CORRECTION AS HOLONOMY

The correction ledger is the retained state that lets the programme return to an old question without pretending it has never been there. A project that preserves only conclusions has a path-dependent self-image; one that preserves corrections can close more of the loop.

What a typed remainder protects

Remainder	Meaning	Why it cannot be silently reused
[R]_transport	Information lost or unverified during a declared map	Not automatically a topological obstruction

Remainder	Meaning	Why it cannot be silently reused
[R]_coh	Measured triple-overlap or matrix coherence defect	Not a cohomology class without a site, coefficients, and coboundary
[R]_id	Gap between one operation and one self-model	Zero closes one instrument, not complete self-description
[R]_finite	Finite-size or finite-dimensional limitation	Cannot support an asymptotic or infinite-dimensional conclusion
[R]_witness	No independent recomputation or external witness	Valid provenance still does not create independence

Repository source trail: src/aionq/movingframe.py; src/aionq/stack.py; docs/ARCHITECTURE.md revision IV; docs/SOURCES_DIGEST.md

PART III

HB-0 as an executable architecture

Worlds, translations, identity, recognition, geometry, and decision without premature collapse

observe | verify | leave the remainder visible

CHAPTER 07

Worldlets: local worlds before global vectors

A Worldlet is a typed local context - not a miniature universe, not a neuron, and not an autonomous mind.

CHAPTER ABSTRACT

This chapter introduces the repository's central custom object from first principles. A Worldlet keeps state, geometry, timing, viewpoint, evidence, and unresolved remainder together long enough to prevent premature flattening. The chapter distinguishes the abstract Worldlet type from the configurable 32-Worldlet reference factory, explains how the kernel assigns base points, and gives a worked example in which somatic, episodic, social, and prospective readings of one event remain related without becoming identical.

Why a Worldlet exists

Most software systems seek a common representation early. Sensor values, memories, other-agent models, plans, confidence estimates, and error signals are projected into one vector so that a single update rule can act on them. That move is convenient, but it silently assumes that every component already shares a carrier, metric, clock, meaning, and evidential status.

HypercomplexBrain begins from the opposite engineering suspicion: the common vector may be the conclusion of a successful translation, not the starting point. A Worldlet is the container used to delay that conclusion. It preserves a local "way the situation exists for this subsystem" until a declared map earns comparison with another local world.

WHAT A WORLDLET IS NOT

It is not a biological neuron, a conscious subject, an independent software agent, a simulated universe, or a claim that cognition literally decomposes into four named regions. It is a typed architecture object whose value must be established by clearer invariants or better held-out performance than simpler representations.

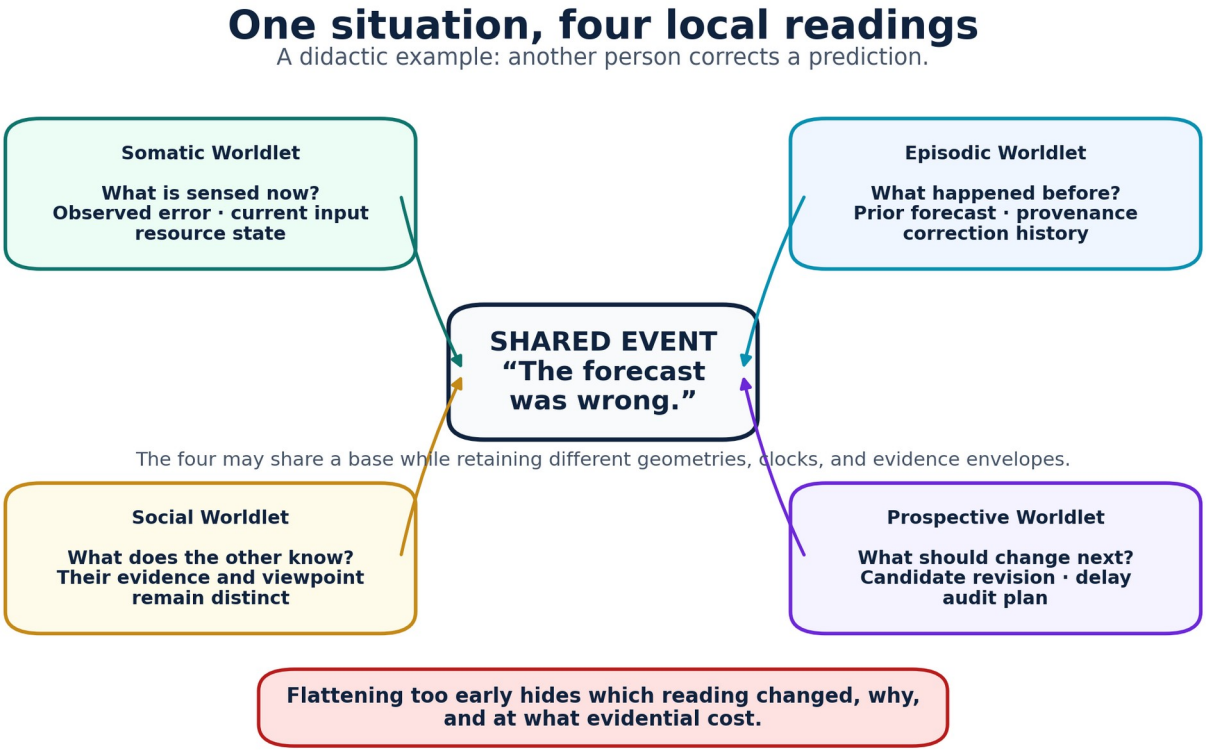


Figure 7.1. A didactic four-Worldlet reading of one corrective event.

The formal object, read from inside out

The architecture document writes a Worldlet as a structured tuple. The notation is compact, but each component has a practical refusal role: it prevents one kind of accidental comparison or overclaim.

$$W_i = (X_i, g_i, A_i, \Theta_i, O_i, \text{boundary}_i, \text{energy}_i, Q_i, R_i)$$

In the executable kernel, the WorldletState object stores the local fields. The PolycosmicKernel separately assigns every Worldlet exactly one immutable point b_i on the shared base B. This distinction matters editorially: the base point is a kernel-level contract, not a field that the Worldlet can freely rewrite.

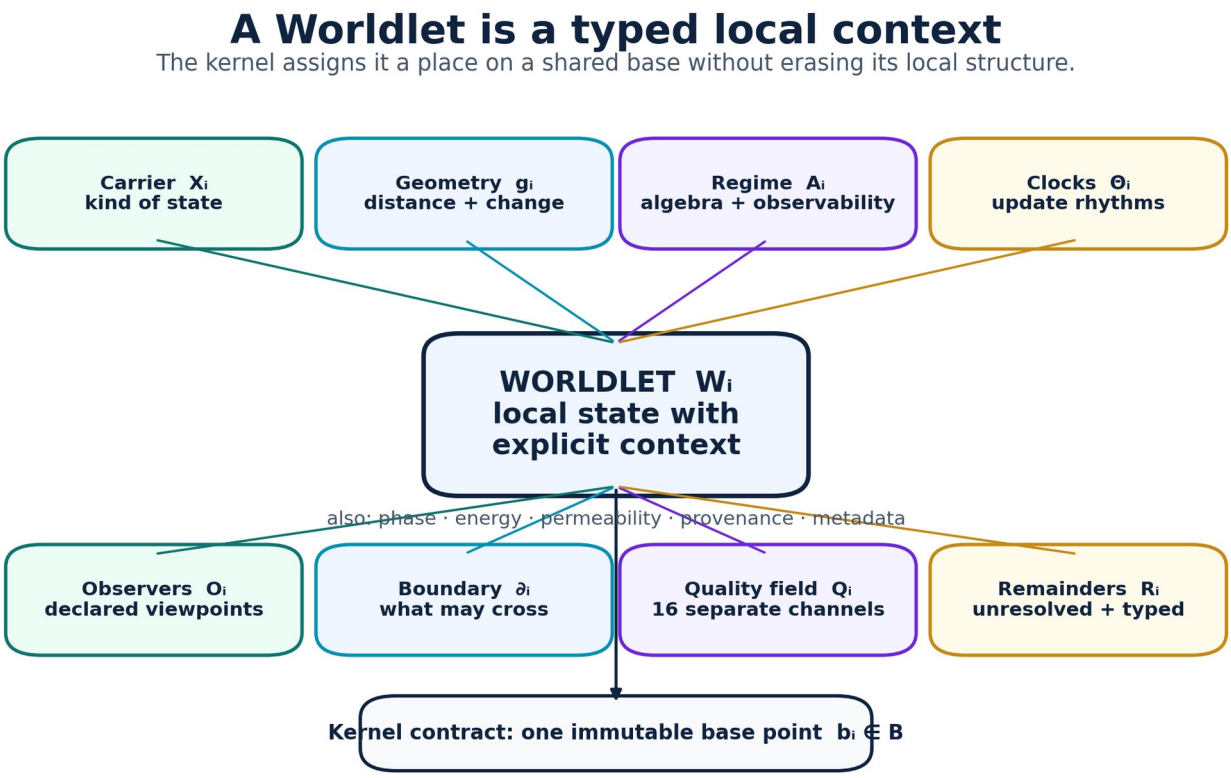


Figure 7.2. The Worldlet anatomy and the separate kernel-level base-point contract.

Field	Plain meaning	Executable surface	Why it remains separate
X_i / carrier	The kind and dimension of local state.	carrier + coordinates	Equal array shape does not establish a common semantic carrier.
g_i / geometry	How distance and local change are measured.	local_geometry + optional metric_tensor	Poincaré, Fisher, Riemannian, Euclidean, and discrete updates obey different rules.
A_i / regime	The algebraic or observability convention.	algebra_regime	A quaternion label or real control is diagnostic, not interchangeable by name.
Theta_i / clocks	Which rhythms may update the state.	clocks referencing ClockSpec objects	Fast inference and slow identity/evidence retention should not drift at one rate.
O_i / observers	Declared viewpoints or audit roles.	observers	Observation is part of the model, not a view from nowhere.
boundary_i	What can enter or leave this local context.	boundary + permeability	A local context needs a translation gate rather than unrestricted absorption.
energy_i / phase	Local resource and phase telemetry.	energy + phase	These are engineering state variables, not physical-energy or consciousness claims.
Q_i	Sixteen separate quality channels.	MindQualityField	The implementation refuses a weighted-sum promotion scalar.
R_i	Typed unresolved objects.	RemainderEnvelope tuple	Semantic, transport, identity, coherence, empirical, and

Field	Plain meaning	Executable surface	Why it remains separate
			formal gaps cannot be merged silently.
b_i in B	The Worldlet's location in the universal atlas.	PolycosmicKernel.base_points	Shared location does not collapse local fiber geometry.

One base, many local fibers

Two distinctions must be kept apart. First, Worldlets have positions on one shared Poincaré base so the architecture can speak about relative placement. Second, their local state spaces remain heterogeneous fibers. A somatic Worldlet can use a Poincaré-ball update while an episodic Worldlet uses a Fisher metric and a prospective Worldlet uses an explicit Riemannian chart.

The shared base therefore behaves like a common atlas for locating local worlds, not like a universal feature vector that makes every local coordinate directly comparable. WorldletState.distance_to refuses carrier, dimension, geometry, or metric mismatches. The required next operation is transport through a declared connection, not a convenient subtraction.

The reference factory is an example, not a law

The default build_hb0() factory constructs thirty-two Worldlets: eight in each of four regimes. The implementation is more general than that picture. It accepts any number that is a multiple of four and at least eight, and it allows the local fiber and base dimensions to be chosen subject to explicit checks. Thirty-two is therefore a reference specimen for tests and discussion, not a biological constant or a theorem about the correct number of cognitive contexts.

Reference regime	Default local geometry	Engineering question	Two default clocks
Somatic	Poincaré ball	What is presently embodied or sensed?	fast + meso
Episodic	Fisher metric	What history, provenance, or belief state matters?	meso + slow
Social	Euclidean control	What model of another viewpoint is being kept distinct?	fast + meso
Prospective	Riemannian chart	What future transition or audit is being considered?	meso + slow

These labels are architecture roles. They do not assert that the implementation possesses a body, autobiographical memory, social understanding, or foresight. Such claims would require external tasks and controls that the reference factory does not provide.

A worked example: correction without flattening

Suppose the system predicts that an experiment will succeed, and a collaborator returns evidence that the prediction was wrong. A one-vector architecture might overwrite a global state with a

corrective gradient. The Worldlet architecture asks what kind of change occurred before deciding how to integrate it.

1. The somatic Worldlet records the present signal: the observed failure, resource state, and immediate input conditions.
2. The episodic Worldlet retains the original prediction, the evidence that supported it, the run configuration, and prior corrections.
3. The social Worldlet models the collaborator's evidence and perspective without replacing it with the system's own summary.
4. The prospective Worldlet proposes a revised experiment, delayed checks, and consequences for future claims.
5. Tendrils translate only the components required for coordination and report what was preserved, transformed, or lost.
6. The governor accepts the integrated update only if coherence, identity, non-absorption, compute, plurality, anti-Löb, and fragmentation gates remain viable.

EDITORIAL READING RULE

When the word "Worldlet" appears, translate it first as "typed local context with its own update semantics." Only then ask whether a more ambitious cognitive interpretation has been earned.

Lifecycle and failure conditions

Stage	What happens	Failure that should remain visible
Create	Declare carrier, geometry, clocks, observers, boundary, qualities, remainders, and kernel base point.	Missing type, invalid metric, out-of-ball coordinate, or undeclared clock.
Update locally	Apply a geometry-aware delta only when one of the Worldlet's clocks is due.	Illegal discrete update, non-finite state, or chart exit.
Translate	Use a Tendril rather than comparing heterogeneous arrays directly.	Carrier mismatch, absent return map, high defect, loss, or excess cost.
Integrate	Evaluate transport, higher coherence, identity, and vector viability gates.	Rejected transition restores pre-step coordinates.
Retain / deweave / bud	Keep useful correspondences; record deweaving; permit morphogenesis only under a stronger benchmark contract.	Decorative growth without a persistent typed obstruction or matched-width gain.

Evidence status and the simplest kill test

At evidence level S/H, the Worldlet is an executable architecture hypothesis. The code proves useful local type and refusal properties about its own objects, but it does not show that a Worldlet decomposition improves cognition or even prediction. The smallest empirical kill test is straightforward: compare the full local-world representation against a capacity- and compute-matched typed baseline on tasks that genuinely require context-dependent geometry, timing, and provenance. If the simpler representation matches held-out competence and correction retention, the extra ontology should be simplified or retired rather than defended by richer language.

Repository source trail: docs/HB0_POLYCOSMIC_KERNEL.md; src/aionq/hb0/core.py; src/aionq/hb0/kernel.py; tests/test_hb0.py

CHAPTER 08

Tendrils: translation with a receipt

A Tendril relates local worlds by declaring maps, losses, costs, and an unresolved remainder.

CHAPTER ABSTRACT

This chapter explains why a Tendril is stricter than a message bus or embedding layer. It may have a forward map and an explicit return map, or it may report that no return exists. Its authoritative output is a raw transport defect plus typed preservation/loss metadata; the convenience fidelity exp(-defect) never replaces that report. A worked dimensionality-reduction example shows how a low-looking score can still hide an irrecoverable distinction.

Why translation is a first-class object

Once Worldlets are allowed to keep different carriers and geometries, communication cannot be treated as copying bytes between equivalent containers. A translation may rotate coordinates, change dimension, alter phase, collapse a null space, or preserve only a task-specific statistic. The architecture must therefore represent the translation itself and not only its output.

$$T_{ij} = (F_{ij}, F_{ji}?, \text{preserved}, \text{transformed}, \text{lost}, \text{cost}, [R]_{\text{transport}})$$

The question mark is important. In the implementation, the backward map is optional. When it is absent, the ChartFunctor must state why return is unavailable. The Tendril then emits an H-level transport remainder and zero convenience fidelity instead of pretending that one-way projection is reversible.

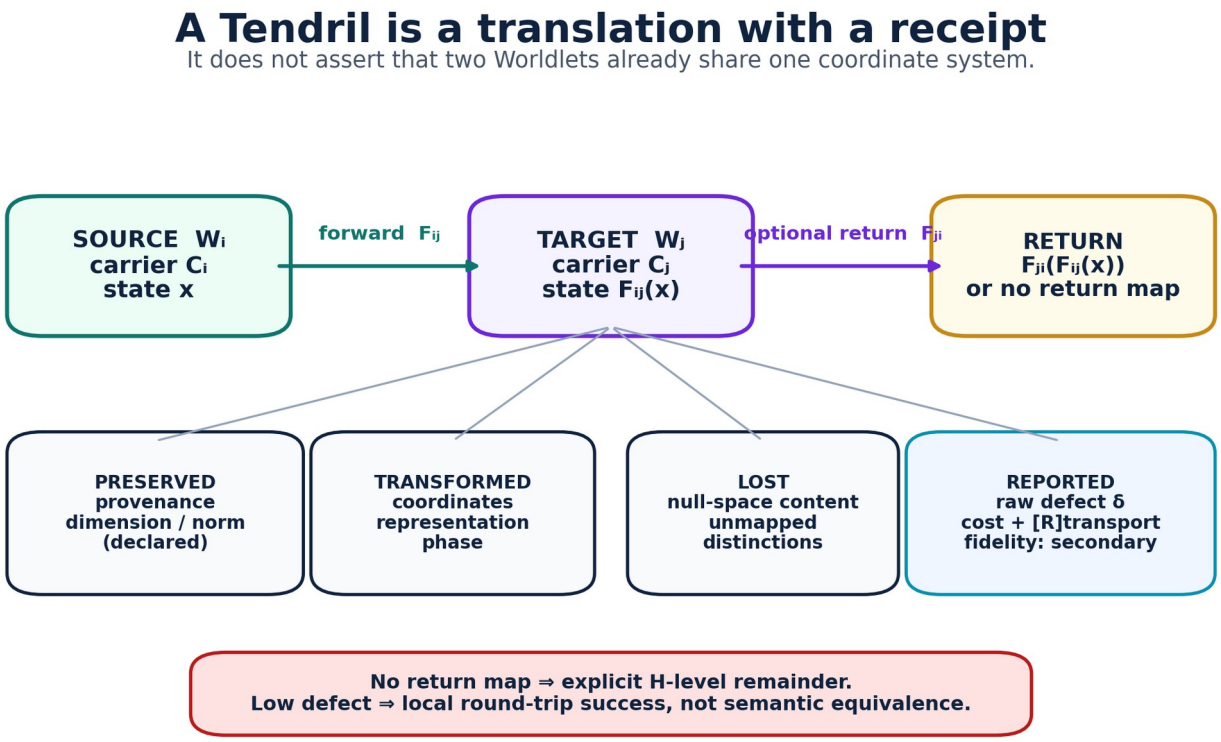


Figure 8.1. A Tendril exposes the translation path and its receipt.

The receipt fields

Receipt field	Question answered	What it prevents
Forward map F_ij	How is source state represented in the target carrier?	Implicit casting or subtraction across incompatible carriers.
Return map F_ji	Can the translated state be brought back to the source carrier?	Calling an encoder reversible when no decoder is declared.
Preserved	Which invariants are claimed to survive?	A vague “same information” claim.
Transformed	Which coordinates or conventions legitimately change?	Treating a coordinate change as evidence loss or gain.
Lost	Which distinctions are known not to return?	Hiding null spaces or lossy summarization.
Cost	What resource is charged for the relation?	Free communication inside a resource-governed architecture.
[R]_transport	What unresolved defect remains, under which instrument?	Reusing a transport error as a semantic or cohomological remainder.

Raw defect before convenience fidelity

$$\text{delta_ij}(x) = ||\, F_ji(F_ij(x)) - x \,||_2 \quad \text{and} \quad \text{fidelity} = \exp(-\text{delta_ij}(x))$$

The exponential score is useful for dashboards because it maps a non-negative defect into [0,1]. It is not a proof of equivalence, and it is not the primary measurement. The raw defect remains attached

to its carrier and instrument. Two defect values are comparable only when the channel, carrier, map, and norm agree.

Worked example: an apparently good translation that loses one distinction

Let a source Worldlet hold x = (0.6, 0.2, 0.1). A target context requires only two coordinates, so the forward map keeps the first two: $F_{ij}(x) = (0.6, 0.2)$. The declared return inserts zero in the omitted coordinate: $F_{ji}(F_{ij}(x)) = (0.6, 0.2, 0)$.

$$\delta_{ij}(x) = 0.1 \qquad \text{and} \qquad \exp(-\delta_{ij}(x)) \approx 0.905$$

A dashboard could display 0.905 and tempt the reader to say that translation is “ninety per cent faithful.” The receipt gives the more useful conclusion: the third distinction is in the null space and cannot be recovered. Whether that loss is acceptable depends on the task. The score cannot answer that question by itself.

SEMANTIC CEILING

A low round-trip defect establishes local reconstruction under the declared maps and norm. It does not establish that the source and target concepts mean the same thing, that every relevant invariant was preserved, or that the map generalizes beyond the measured states.

Weaving and deweaving

A Tendril remains woven only while return, coherence, and resource gates hold. Deweaving is not deletion. The relation records the reason it was disabled, preserving the correction history. This matters because a future representation may repair the translation, and because silently dropping a failed relation would make the system's history falsely coherent.

Reason to deweave	Observed signal	Recorded consequence
Return failure	Round-trip defect exceeds the frozen gate.	Disable the relation and keep [R]_transport.
Unstated loss	A supposed invariant changes or a null space appears.	Correct the receipt; do not rescue the old claim with a new score.
Coherence failure	Composed maps disagree on a declared overlap.	Keep [R]_coh and inspect the maps or scope.
Resource violation	Transport cost breaks a component budget.	Reject the transition; another credit cannot cancel it.
Absorption	Target similarity rises because source distinctions are erased.	Recognition/resonance promotion is killed.

At evidence level P for the implementation's local refusal properties and M for a particular measured defect, the Tendril is a useful transport instrument. The broader claim that Tendril-mediated composition improves competence remains H and requires matched baselines.

Repository source trail: src/aionq/hb0/core.py (ChartFunctor, ProfunctorTendril, TransportObservation); docs/HB0_POLYCOSMIC_KERNEL.md

CHAPTER 09

CosmosCages and GerbeCells: scope before global coherence

Nested contexts and triple-overlap witnesses are implemented as structural tests, not as completed topology.

CHAPTER ABSTRACT

This chapter separates two often-confused ideas. CosmosCages define an acyclic hierarchy of local scope. GerbeCells measure a supplied matrix composition around a declared triple. Both are valuable because they can refuse malformed structure, but neither automatically supplies a mathematical cover, a site, a sheaf, a gerbe, or a cohomology class. The chapter states exactly what the current code checks and what additional objects would be required for the stronger interpretation.

CosmosCage: a scope tree, not a physical cosmos

The term “CosmosCage” names a Russian-doll scope container. A root cage can contain child cages; each cage names a local geometry, a base identity, and Worldlet IDs that are considered in scope. The nesting gives the architecture a place to ask local-versus-global questions and a place where future morphogenesis could occur.

The present add_child implementation performs two exact checks: the cage graph must remain acyclic, and cage IDs must remain unique across the incoming subtree. The stronger phrase “no duplicate containment” would overstate the code. The class does not by itself prove that a Worldlet appears in only one cage, that the cages form a topological cover, or that intersections obey cover axioms.

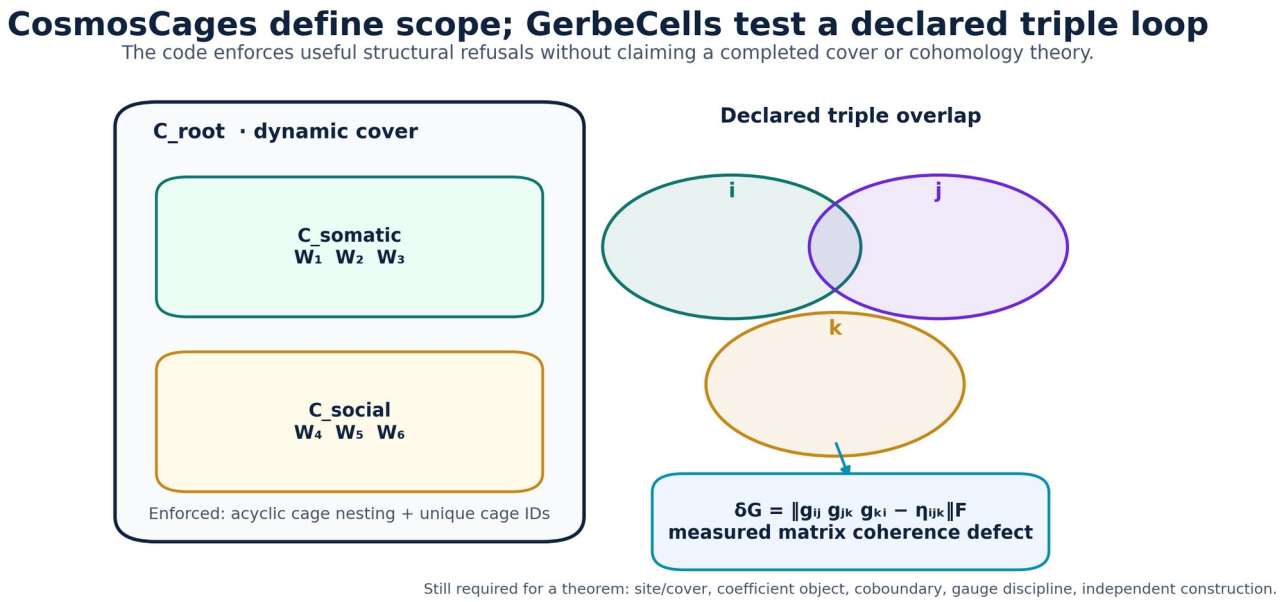


Figure 9.1. Current scope and coherence objects, with their theorem boundary.

Object	Current executable guarantee	Not yet guaranteed
CosmosCage	Acyclic child nesting; no duplicate cage IDs in the joined tree; recursive lookup path.	Unique Worldlet membership, cover completeness, overlap axioms, dynamic morphogenesis theorem.
Kernel cage audit	Every Worldlet appears somewhere in the walked cage hierarchy; every cage carries the shared base ID.	Minimality, disjointness, or a canonical cover selected by data.
GerbeCell	Equal-size square matrices; Frobenius defect of g_ij g_jk g_ki against supplied eta_ijk.	A site, coefficient object, transition maps derived from data, gauge invariance, coboundary operator, or cohomology class.

GerbeCell: a triple-overlap witness

$$\text{delta_G} = || \; g_ij \; g_jk \; g_ki - \text{eta_ijk} \; ||_F$$

The measurement asks whether three supplied transition matrices compose to the supplied comparison matrix eta_ijk. A zero defect means that this finite matrix equation closes under the chosen data. A non-zero defect records [R]_coh with its carrier and instrument.

The word “gerbe” points toward a richer mathematical ambition, but the executable object is deliberately narrower. To promote it into a gerbe or cocycle statement, the programme would need at least a specified site or cover, a coefficient object, transition data with a defined provenance, a coboundary operator, an equivalence/gauge discipline, and a proof that the matrices instantiate those objects. Without that bridge, the correct phrase is “triple-overlap matrix coherence witness.”

WHY THE NARROW WORDING IS STRONGER

A measured defect that is correctly typed can guide debugging and future formalization. A prematurely promoted cohomology label creates authority without supplying the missing structure. The narrower object is easier to falsify and therefore more scientifically useful.

Worked reading: zero is not automatically global coherence

In the reference factory, the two GerbeCells are initialized with identity matrices. Their defect is therefore exactly zero by construction. That value is a useful integration check: the matrix plumbing is consistent. It is not evidence that a learned or externally derived cover has non-trivial higher coherence. A future experiment must construct transition data independently of the equation used to score it, or the witness and its oracle remain correlated.

At evidence level P for the finite matrix calculation and S/H for the intended geometric interpretation, the current objects are credible software scaffolds. Their most important peer-review obligation is source independence: the same narrative should not choose the overlap, construct the matrices, choose eta, and then promote the resulting defect without a matched null or an independent oracle.

Repository source trail: src/aionq/hb0/core.py (CosmosCage, GerbeCell); src/aionq/hb0/kernel.py; docs/HB0_POLYCOSMIC_KERNEL.md

CHAPTER 10

Polyholonomic identity and recognition without absorption

Identity is treated as consistency among local regimes, while recognition must preserve the difference of the other.

CHAPTER ABSTRACT

This chapter translates two of the repository's most philosophical terms into bounded software objects. Polyholonomic identity consists of local sections, local holonomies, comparison maps, higher-coherence witnesses, a viability region, and a typed identity remainder. Recognition is directional and rewards useful gluing plus preserved difference while penalizing absorption and projection violence. Neither construct is evidence of a self, consciousness, empathy, or social understanding; each is an instrument that can be tested and retired.

Why identity is not one vector

A global identity vector is easy to compare over time but tends to erase the fact that a system behaves differently in bodily, episodic, social, and prospective regimes. HypercomplexBrain instead models an identity ecology. Each regime has a local section and one or more local holonomy matrices. Declared comparison maps carry sections between regimes and back. Higher cells supply an additional finite coherence check.

$$I = (\{I_{alpha}\}, \{J_{alpha,beta}\}, \{G_{alpha,beta,gamma}\}, V_I, [R]_{id})$$

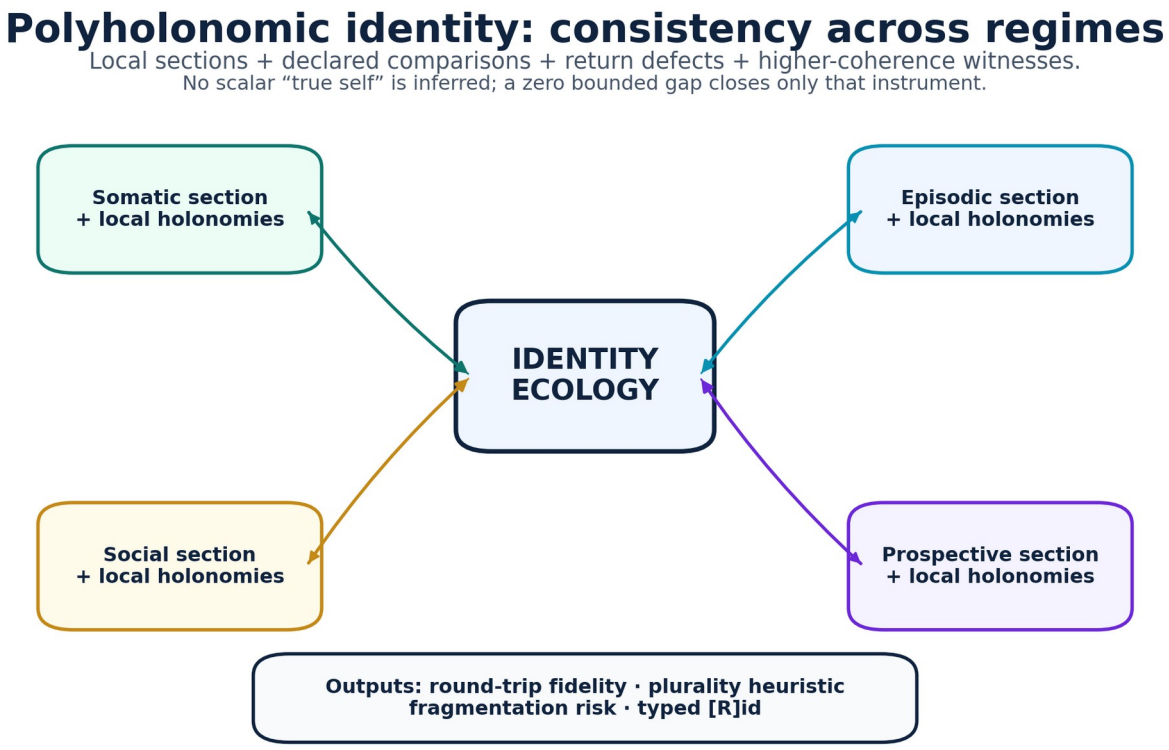


Figure 10.1. Identity as an ecology of local sections and declared comparisons.

The implementation exposes three principal summaries. Identity fidelity averages exp(-defect) over declared round trips. Plurality estimates the normalized effective rank of the regime sections, but only when they already share one section dimension. Fragmentation risk combines missing comparison links, round-trip defect, and higher-coherence defect. These are engineered heuristics, not validated psychological measurements.

Instrument	What it computes	Interpretation ceiling
Round-trip identity fidelity	Mean exp(-||J_back(J_forward(section)) - section||).	Local return consistency under declared comparison maps.
Plurality	Normalized effective rank of regime sections in a common dimension.	Diversity of the current section matrix, not richness of a person or agent.
Fragmentation risk	Average of connectivity gap, transport risk, and coherence risk.	A bounded architecture heuristic whose weights are not empirically validated.
Gödelian [R]_id	L2 gap between one operation vector and one self-model vector.	A bounded self-model instrument; not Gödel’s theorem and not complete self-description.

Recognition is directional

$$\mathrm{Rec}(A \to B) = [\ \mathrm{gluing} + \mathrm{useful_update} + \mathrm{preserved_difference} - \lambda_A\ \mathrm{absorption} - \lambda_V\ \mathrm{projection_violence}\ ]_+$$

A model of B can be useful to A while still erasing what makes B different. The directional score therefore includes preserved difference as a positive term and absorption as a penalty. Mutual recognition uses both directional scores and discounts viewpoint-swap defect. Resonance additionally depends on phase, distance, gluing defect, return fidelity, telos alignment, and a non-absorption factor.

Because the resonance expression includes a cosine of phase difference, it can be negative. That sign is a mathematical feature of the chosen formula, not a clinical or moral classification. The formula remains a design hypothesis until a task shows that it predicts better coordination or safer correction than simpler similarity and calibration metrics.

Worked contrast: agreement versus absorption

Case	Surface similarity	Preserved difference	Absorption	Authorized reading
A copies B's conclusion and deletes B's evidence provenance.	High	Low	High	Similarity is not recognition; the score should be penalized.
A updates its plan using B's evidence while retaining source, uncertainty, and disagreement.	Possibly moderate	High	Low	A candidate instance of non-absorptive useful recognition.
A and B both return through their declared maps with low defect, but the task outcome does not improve.	High local consistency	Declared	Low	Transport works; competence benefit remains unestablished.

NON-SENTIENCE BOUNDARY

"Identity," "recognition," "other," and "resonance" denote software structures and engineered observables in this repository. They do not establish subjectivity, consciousness, empathy, personhood, or AGI.

The smallest empirical test is a coordination problem with private information: compare the full recognition coupler against matched baselines that optimize task reward, similarity, or calibration. Promotion requires better held-out coordination while preserving provenance and non-absorption gates. If those gates add no discriminatory value, the terminology should contract to the simpler operator that survives.

Repository source trail: src/aionq/hb0/identity.py; src/aionq/hb0/core.py; docs/HB0_POLYCOSMIC_KERNEL.md

CHAPTER 11

Sixteen engineering qualities, vector governance, and plural time

The “mind qualities” are independent instrumentation channels; the implementation refuses to collapse them into one reward.

CHAPTER ABSTRACT

This chapter explains the sixteen quality coordinates, their evidence obligations, the componentwise Hamiltonian envelope, and the three default clocks. The vocabulary is intentionally aspirational, but each term terminates in an executable score, a risk, an obligation, or a kill condition. The quality vector is not psychometrics and not evidence of a mind. Its strongest current property is anti-flattening: weighted_sum() raises an error, so a gain in exploration cannot cancel an identity, absorption, or remainder-stewardship failure.

Why sixteen channels instead of one score

A scalar reward creates exchange rates among unlike failures. A system can gain enough task reward to hide provenance loss, enough coherence to hide absorption, or enough exploration to hide failed return. HB-0 rejects that arithmetic. MindQualityField stores sixteen values in [0,1], compares them by componentwise floors or Pareto relations, and raises FlatCollapseError when asked for a weighted sum.

Core channel	Conjugate channel	Engineering reading
Grounding	Embodiment	Bind a state to evidence and resource headroom.
Self-reflection	Temporal polyphony	Compare operation with a bounded self-model across distinct clocks.
Gödelian identity	Contextual composition	Retain a typed non-closure channel and measure associator/gluing defects.
Other recognition	Local-global literacy	Model another context while preserving difference and understanding transport limits.
Mutual resonance	Holoportive reconstruction	Require non-absorption, return fidelity, and explicit null-space loss.
Adiabatic patience	Metaplasticity	Separate fast state adaptation from slow identity/rule change.
Ergocetic exploration	Judicial correction	Trade coverage against compute/risk and retain corrections.
Erdodetic return	Remainder stewardship	Bring discoveries back into tests, integration, retirement, and typed

Core channel	Conjugate channel	Engineering reading
		unresolved channels.

The names are not standard scientific constructs. Their legitimacy depends on the operator underneath them. For example, grounding is an evidence-tag strength multiplied by whether references exist; self-reflection is exp(-self-model gap); temporal polyphony combines the number of clocks with desynchronization; judicial correction counts retained corrections; remainder stewardship counts typed remainders. Those definitions make the vocabulary inspectable, but they do not validate it as a model of cognition.

Obligations and kill conditions

Channel	Required evidence obligation	Named kill condition
Grounding	Attach a source, artifact, or explicit S/H status.	Do not promote unreferenced fluency as evidence.
Gödelian identity	Retain [R]_id without promoting it to cohomology.	A zero local gap must not become complete self-description.
Other recognition	Measure viewpoint swap and preserved difference.	Kill if apparent agreement is produced by absorption.
Ergocetic exploration	Return discoveries to controls and provenance.	Kill if added coverage is purchased by hidden compute or coherence loss.
Holoportive reconstruction	Report null-space information loss.	Kill perfect-reconstruction language when boundary data are non-injective.
Metaplasticity	Measure held-out gain after rule change.	Kill if rule change adds no held-out benefit.

A vector Hamiltonian, not a universal energy

$$H = (H_{compute}, H_{coherence}, H_{identity}, H_{absorption}, C_{explore}, C_{integrate})$$

The governor checks component budgets and a viability region. A transition can be rejected for excessive gluing defect, identity defect, absorption, compute, Löbian self-endorsement risk, fragmentation, or insufficient plurality. Exploration credit cannot pay an unrelated identity debt. Rejected transitions restore the pre-step Worldlet coordinates.

The term “Hamiltonian” should be read as contact/symplectic-inspired engineering structure. The implementation does not claim literal energy conservation in a learning system. It supplies a vector envelope and a constrained update rule.

Plural time

Clock	Period / rate	Default role	Editorial interpretation
fast	1 / 1.00	Local inference	Changes on every tick; immediate adaptation.

Clock	Period / rate	Default role	Editorial interpretation
meso	4 / 0.10	Coalition consolidation	Integrates local changes less frequently and at lower rate.
slow	16 / 0.01	Identity and evidence	Protects retained identity/evidence from changing as quickly as inference.

A Worldlet updates only when one of its declared clocks is due, and it uses the minimum active rate. The model therefore makes temporal separation executable. Whether periods 1, 4, and 16 are useful is an empirical question; they are defaults, not universal cognitive constants.

BORDÓN RESPIRATION

The gaita-inspired reservoir separates steady invariant funding from speculative draw. Buffer capacity cannot fund an indefinitely uninterrupted channel. The metaphor is valuable only because it terminates in a resource ledger and a failure mode.

At evidence level P for the data-structure and refusal semantics, S/H for the cognitive vocabulary, and M only when a particular operator is run on specified data, the sixteen-channel field is a governance instrument. Its next obligation is calibration against simpler safety and performance measures on held-out tasks.

Repository source trail: src/aionq/hb0/mind.py; src/aionq/hb0/core.py; src/aionq/hb0/kernel.py; docs/HB0_POLYCOSMIC_KERNEL.md

CHAPTER 12

The shared Poincare H^4 base

A declared coordinate contract, not a measurement that every hierarchy is naturally hyperbolic.

HB-0 places every Worldlet on one shared Poincare base H^4. The word “universal” means unique within one architecture instance. It does not mean that every local object has the same metric, or that the free-theory graph empirically generated negative curvature.

This distinction exists because an earlier argument failed. The five-node free-theory containment graph is K(2,3), with normalized hyperbolicity delta/diameter = 0.5. That measurement did not justify a tree-like hyperbolic base. The repository preserved the negative result and later reinstated H^4 explicitly as an A-tagged architectural axiom. The base is a design contract, not a retroactive positive measurement.

The mathematical implementation uses the future hyperboloid in R^(1,4) together with the open Poincare ball as an isometric chart. It provides distance, exponential and logarithm maps, Levi-Civita parallel transport, Lorentz isometries of the hyperboloid realization, and fail-closed numerical checks near the floating-point boundary.

HUMAN TRANSLATION

The shared base is a map sheet on which local worlds are located. Choosing a map is useful even when the road network did not prove that the map is uniquely natural. The A tag keeps that choice honest.

Local geometry remains heterogeneous

Worldlets may use Poincare, Fisher, ordinary Riemannian, Euclidean-control, quaternionic, split-signature, or positive-semidefinite observation structures in their fibers. These structures are typed separately. The ambient Lorentz form used to realize the hyperboloid is not the same object as the independent split (2,2) carrier elsewhere in the repo. An isotropic vector in a split form is not automatically in the kernel of a PSD observation operator.

The reference cover deliberately mixes local geometries. That heterogeneity is part of the experiment: a universal base should coordinate locations without erasing the local representation most appropriate to each world.

The Euclidean ablation

A tangent-matched Euclidean control uses the same dimension and local scale at the origin. This makes curvature the intended difference rather than capacity or units. The control is not allowed to “vote” the base away; it is an ablation used to test whether curvature contributes to a later task under matched conditions.

At present, the universal H^4 base remains an architectural A claim. It has structural tests and a provenance envelope, but the repository's own review notes that the envelope is self-recomputed, unsigned, and not an independent empirical selection of hyperbolic geometry.

EVIDENCE BOUNDARY

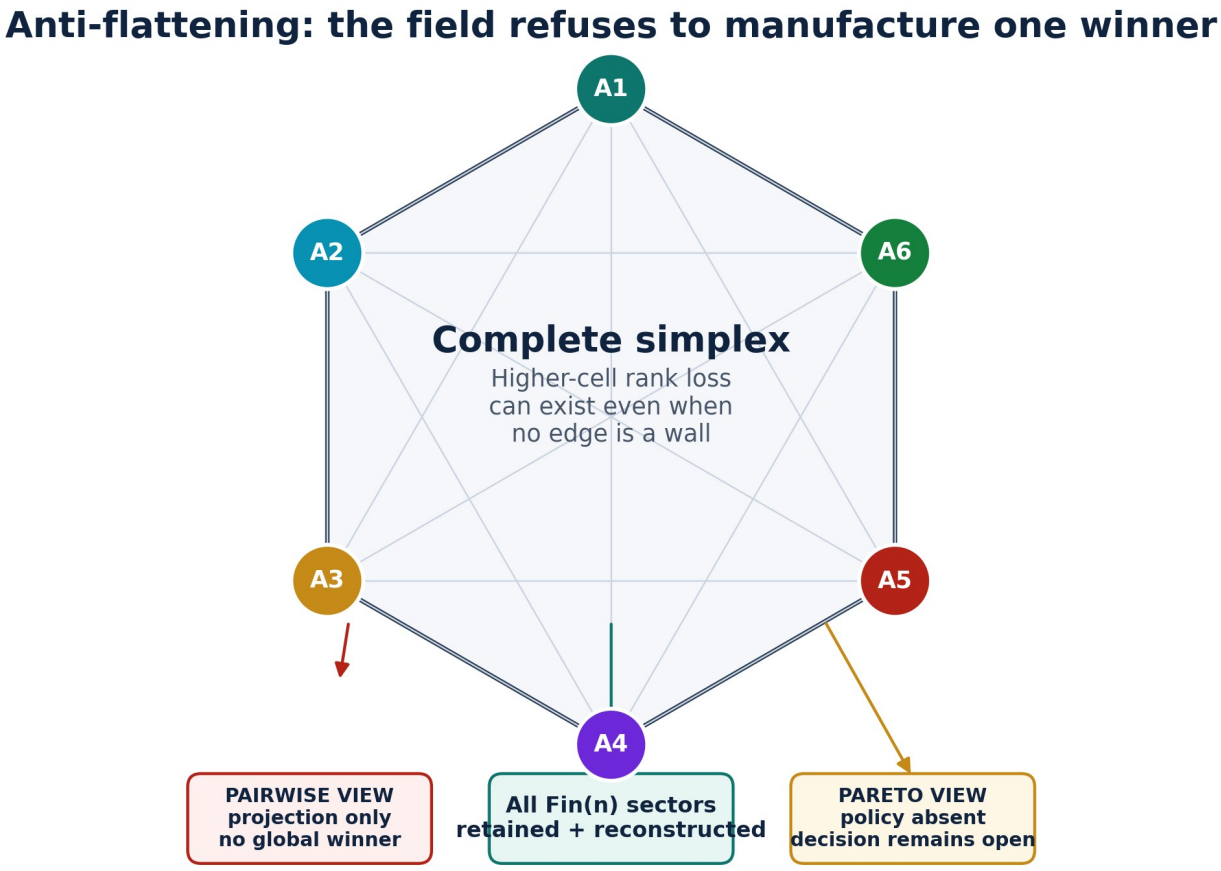
Nothing about the shared base establishes hyperbolic task advantage, biological geometry, cognition, or optimality. Those require a held-out comparison against the tangent-matched Euclidean and other mandatory controls.

Repository source trail: docs/HB0_HYPERBOLIC_BASE.md; docs/REPOSITORY_REVIEW_2026-07-30.md; src/aionq/hyperbolic.py

CHAPTER 13

The n-ary hypercomplex decision field

The field keeps higher-order disagreement visible instead of manufacturing a binary winner.



Pairwise comparisons are projections of a higher-cell object. The global decision remains open until policy and evidence are supplied.

The decision field represents a complete simplex of three to eight alternatives. Each vertex carries a point on the shared base, a quaternion fiber value, the sixteen quality coordinates, a vector Hamiltonian envelope, and a typed evidence descriptor. Quaternion coordinates are not ordered. Benefits, costs, and evidence keep their own partial orders.

The field uses centered contrast spaces to ask how much n-ary structure an observation channel loses. This can reveal a genuine higher-cell wall even when every pairwise edge remains visible. In the repository's example, all three alternatives differ in an observed real coordinate, yet a ternary contrast falls into the observation kernel. A pairwise-only system would miss the obstruction.

For cells with more than five alternatives, a four-real-dimensional quaternion carrier cannot represent every independent contrast. The code emits [R]_intrinsic_cell_collapse rather than compressing the excess and reporting complete resolution.

Hypercomplex structure is load-bearing but bounded

Relations use actual quaternion multiplication, including oriented ternary channels. Common unit-quaternion transport acts by conjugation, and associativity makes chart composition legitimate. Octonions remain a bracketed diagnostic; they are not used as if they supplied associative flat transitions. Sedenions are not treated as invertible transport.

The field also retains exterior or Plucker coordinates for higher cells on the hyperbolic base and all finite Fourier sectors k in Fin(n). Reconstruction and Parseval checks ensure that a mean, argmax, or single sector has not silently replaced the whole family.

Why there is no global winner

Pareto dominance is available as a diagnostic projection. An alternative may be no worse in all quality benefits and no more costly in all Hamiltonian components, with a strict improvement somewhere. Even then, the result is only a Pareto singleton candidate. The global decision flag remains false until a typed n-ary policy and runtime evidence resolver are registered.

This is deliberate incompleteness. The architecture prefers an open multisection to a fake certainty produced by choosing arbitrary weights. It is an executable version of the principle that many real decisions are not reducible to one total order without an explicit political or operational policy.

> **WHAT WOULD COUNT AS PROGRESS**
>
> A legitimate next step is not “add more hypercomplex dimensions.” It is to demonstrate that the retained higher-cell information improves a frozen task against dimension-matched real and pairwise controls, under equal compute and with the same evidence resolver.

Repository source trail: docs/HB0_HYPERCOMPLEX_DECISION_FIELD.md; src/aionq/hb0/decision_field.py; src/aionq/cayley_dickson.py

PART IV

Quantum, algebraic, and information surfaces

Exact structures, bounded simulations, and the costs that prevent metaphor from becoming a claim

observe | verify | leave the remainder visible

CHAPTER 14

Quantum fibers and the price of noncompact depth

The repository separates a geometric substrate from the compact quantum operations that can actually be implemented.

QSigma-R6: depth is the bill

The QSigma-R6 round asks a precise representation question. A split-signature substrate can be written as SL(2,R). Quantum computation on finite systems is unitary, while noncompact directions do not admit a faithful finite-dimensional unitary realization. The experiment therefore prices a non-unitary block-encoding route rather than claiming that hyperbolic motion is already a gate.

In the chosen coordinates, the singular values are exactly exp(rho) and exp(-rho). The element is orthogonal - and therefore directly unitary in the real representation - if and only if rho = 0. The compact angular coordinates are free; the radial hyperbolic depth is precisely what incurs condition number and post-selection cost.

Four preregistered predictions survived in the committed round: depth controls the bill, the elliptic fraction thins with depth, average and worst-case post-selection costs separate, and a closed word whose net product returns to the compact subgroup may be compiled freely only when intermediate states need not materialize. The first numerical run was voided by a strict amplitude control, which caught square-root amplification of floating-point noise at zero depth; the corrected implementation retained that failure history.

THE UNCOMFORTABLE RESULT

Hyperbolic geometry is attractive because depth can carry capacity. R6 finds that the same depth is exactly the coordinate the finite unitary substrate refuses for free. The round neither vindicates nor refutes hyperbolic memory; it names the cost that any real design must pay.

Operational information: compactified, not erased

A later operational analysis normalizes the noncompact matrix and maps depth to q = exp(-2 rho) in [0,1]. The radial ray is not deleted; it becomes exponentially ill-conditioned. Radial Fisher information decays like exp(-4 rho), while two angular directions saturate. The operational space approaches a two-torus boundary with finite Fisher volume.

This distinction is important. Saying “the depth disappears” would be false. Its information remains encoded in a determinant-like coordinate, but recovering it becomes costly. A boundary channel can retain at most a bounded amount of accessible information in the fixed probe even while the source measure grows without bound.

QB1 and QB2: two different quantum shadows

QB1 embeds a point of the ball into a qutrit density with a radial flag and a two-dimensional directional state. It is a typed shadow of the point, not an isometric replacement for H^4. QB2 transports an independent spin-fiber datum: a path in H^4 produces a positive-chiral SU(2)+ holonomy, which can be projected to a one-qubit Qiskit UnitaryGate.

QB2 retains the signed lift, so U and -U are not silently identified by density conjugation. The negative chirality remains an explicit remainder. The Qiskit boundary reconstructs and revalidates the full geometric receipt rather than accepting a detached matrix. Its reports fix simulator_only = true, backend_executed = false, and advantage_claim = false.

EVIDENCE CEILING

These surfaces establish exact identities and bounded numerical consistency for declared models. They do not establish a physical device, noise robustness, channel capacity, speedup, quantum advantage, or cognition.

Repository source trail: docs/QUANTUM_SUBSTRATE_COST.md; docs/OPERATIONAL_INFORMATION.md; docs/HB0_CHIRAL_SPIN_CONNECTION.md; src/aionq/hb0/quantum_base.py; src/aionq/hb0/chiral_spin.py

CHAPTER 15

Memory, reservoirs, process, and visibility

The architecture treats memory as temporal dependence, resource flow, and reconstructability - not as a single stored vector.

Process memory

The process layer distinguishes a state at one time from a process with memory across times. Temporal dependence is measured rather than inferred from recurrent-looking code. This is aligned with the wider architecture: clocks, observers, and transports remain separate inputs, so a system can ask whether an apparent memory effect survives a change of frame or disappears under a Markov control.

The repository's design language of fast, mesoscopic, and slow clocks is therefore not merely scheduling. It creates falsifiable questions about adaptation, consolidation, and identity retention. R1W later tests one such question and shows why a good final-error number is insufficient when the preregistered temporal discriminator does not separate.

Reservoirs and Bordón

The early periodic breathing model forced all channels to stop together. Bordón replaces it with a reservoir. Intake and expenditure can be asynchronous; the invariant drone remains active while other channels spend buffer. The model exposes a conservation law: steady uninterrupted draw cannot exceed steady refill, regardless of reservoir size.

This creates a useful architectural distinction between capacity and sustainability. A large buffer can support a burst or delay failure, but it cannot fund an identity carrier forever. Any cognitive architecture that claims persistent self-maintenance must identify the steady source that pays for it.

Visibility: what bounded local probes cannot see

The visibility instrument asks how much certified global structure remains outside the span of operators visible to probes of depth r. An early attempt equated high Pauli weight with invisibility. A product state immediately killed that idea: a high-weight correlation may be fully determined by one-local data. The corrected instrument uses the rank of locally generated stabilizer structure instead of raw weight.

This is a representative example of the repository's method. A semantic intuition - global structure may be invisible locally - survives, while the obvious metric is retired. The resulting coupling field is an observability instrument, not a theorem that local inference is weak in every architecture.

Self-knowledge and reconstruction

The self-knowledge surface includes both limits and counterexamples. A blanket prohibition on self-tomography was weakened by syndrome extraction: protected logical information can be inferred without destructively learning every microscopic degree of freedom. Holoportive reconstruction likewise reports error and nullity. Perfect agreement on the observed boundary does not reconstruct an unobserved null-space component.

Together these objects support a mature reading of memory and self-modeling: what is knowable depends on the observation channel, coding structure, and task. The architecture records the unseen subspace rather than treating a successful projection as the whole state.

A USEFUL UNIFYING QUESTION

For every memory or self-knowledge claim, ask: which temporal process, which observation channel, which resource budget, and which invisible subspace? The repo's modules are attempts to make those four qualifiers executable.

Repository source trail: src/aionq/process.py; src/aionq/bordon.py; src/aionq/visibility.py; src/aionq/selfknowledge.py; docs/SOURCES_DIGEST.md

CHAPTER 16

Algebra-valued decoding and a killed topological point

The repository's quantum experiments are strongest where they preserve narrow claims and refuse nearby headlines.

S8: the Fano decoder

S8 builds an algebra-valued local decoder in the 28-dimensional coordinate space of real 8 by 8 skew-symmetric matrices. Seven signal generators and twenty-one pairwise commutators form a joint 7 + 21 = 28 system. In the declared linear model, the Fano basis is full rank, nearly orthogonal between signal and crosstalk, well conditioned, and recoverable in the noiseless tests.

The interesting correction concerns comparison. Across one thousand relaxed random-support controls, full crosstalk rank occurred exactly when full signal rank occurred. The original count therefore did not show that Fano uniquely "bought" twenty-one crosstalk modes. A post-hoc conditioning advantage appeared, but because it was discovered after inspecting the rank confound, it remains descriptive until separately preregistered.

S8 quantity	Reported value	Authorized reading
rank(S)	7	Seven declared signal coefficients are identifiable.
rank(C)	21	Twenty-one commutator coefficients are identifiable in this basis.
rank([S C])	28	The full linearized so(8) coordinate system is identifiable.
condition number	2.44949	The chosen basis is well conditioned in the finite model.
Random-support comparison	Rank confounded with signal rank	No unique Fano rank advantage established.

S9: the finite Hofstadter probe

S9 studies one frozen finite interacting Hofstadter system: three hard-core bosons on a periodic 6 by 4 lattice at filling one half. Controls are executed before the target. The target is admitted only if the two lowest levels form a numerically isolated rank-two bundle over boundary-twist space; only then may the determinant line's Chern number be interpreted.

The first run returned a badly non-orthogonal eigenbasis in an essentially degenerate subspace. A separately authorized correction replaced the frame construction with deterministic QR and Rayleigh-Ritz processing. The repair succeeded: the Gram defect fell to numerical precision, neighbor overlaps improved, the raw coarse Chern value remained one, and all thirty-two local U(2) frame attacks on the completed coarse mesh passed.

The parameter point still failed. The bundle width divided by the external gap was approximately 1.0153, above the frozen strict threshold of one. The 9 by 9 confirmation mesh was therefore not admitted. The raw C = 1 remains telemetry; no C/2 response is authorized, and the point is recorded as KILLED_PARAMETER_POINT rather than adjusted until it passes.

WHY THIS NEGATIVE RESULT IS VALUABLE

S9 demonstrates the difference between repairing an instrument and rescuing a hypothesis. The numerical frame was genuinely defective and was fixed. The physical interpretation still failed an independent isolation gate. The repository keeps both facts.

DEGREE DISCIPLINE

The S9 Chern class lives in degree two over a two-dimensional twist torus. It is not interchangeable with flat graph monodromy or arbitrary sheaf H^1. The repository explicitly rejects that biconditional.

Repository source trail: docs/S8_FANO_DECODER.md; docs/S9_FQHE_PROBE.md; src/aionq/fano_decoder.py; src/aionq/fqhe.py

PART V

What the experiments actually say

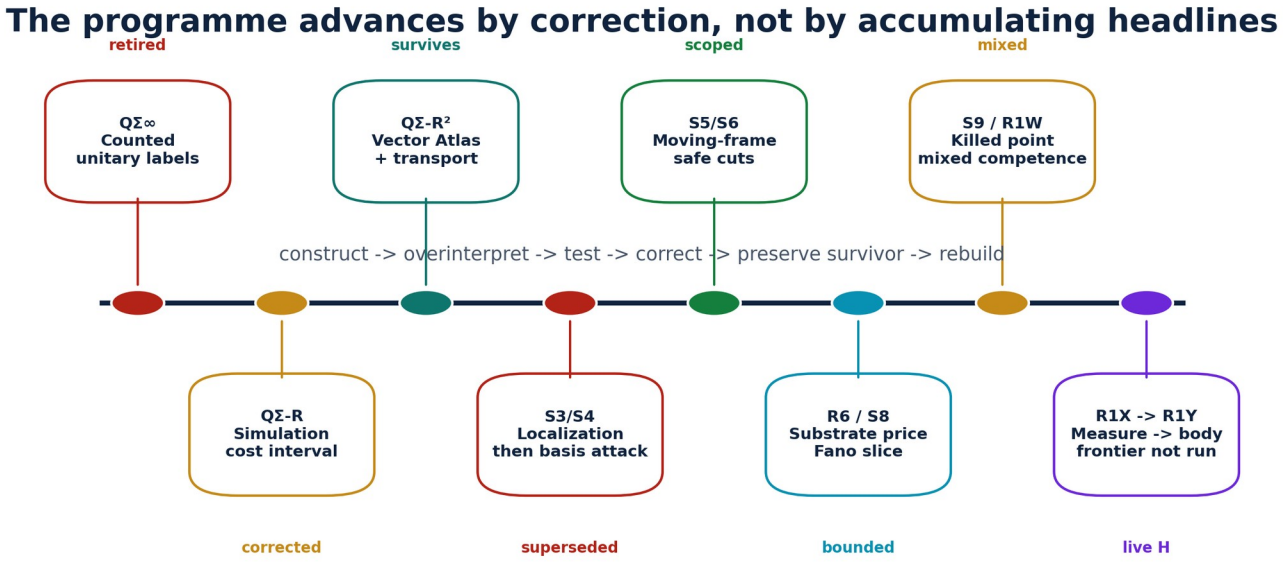
The live knowledge is the surviving floor after controls, corrections, and failed gates

observe | verify | leave the remainder visible

CHAPTER 17

The correction timeline

The programme advances by returning to old questions with a better instrument.



The high-level sequence. Each correction narrows the claim and usually strengthens the research method.

The experiment names can make the repository feel like a long sequence of inventions. A more faithful reading is a sequence of corrections. S1 and S2 establish early novelty and braid or Stokes surfaces. S3 obtains a strong localization result. S4 attacks the representation and shows that most of the effect is basis-relative. S5 and S6 develop moving-frame safe cuts. S7 recomputes external mathematical chains and rejects an overbroad equivalence between curvature, holonomy, and sheaf cohomology. S8 and R6 isolate bounded algebraic and representation results. S9 preserves a killed finite point. The R1 line repeatedly revises how information and partitions are represented.

The same lifecycle appears at several scales: construct, overinterpret, test, falsify or correct, preserve the survivor, rebuild with a sharper type. This is not a rhetorical apology for failure. It is the repository's central discovery process.

S4-G100: a negative result that reorganized the programme

S4-G100 was preregistered with a negative expectation. Six of ten gates failed. The recognition chamber did not beat its deterministic baseline, had severe resource-ratio failures, and was unstable under representation attacks. A component called ordered flow passed its local margin while the full method still lost. A scalar score could have hidden that contradiction; the vector gate prevented it.

The round also changed the interpretation of S3. Under relabelling, the headline sheaf localization AUROC dropped from about 0.789 to 0.551, while matched controls moved much less. The direction survived but only roughly one fifth of the contrast magnitude. The original claim was superseded by a basis-dependent version, not erased.

S5/S6: safe cuts and scoped invariance

The moving-frame response was not to abandon spectral structure, but to stop slicing through degenerate eigenspaces. Safe cuts extend a boundary to the next genuine spectral gap, preserving whole eigenspaces. The reported sweep held the declared invariance across 950 of 950 adversarial instances.

The result remains scoped. Accuracy still depends on graph family and topology. Invariance of the representation does not automatically imply task advantage. The method earns a place as a more trustworthy instrument, not as a universal winner.

Round	Status in the repository	What a human should remember
S3	Superseded	A large localization effect existed in one basis; most magnitude was representation-dependent.
S4-G100	Negative completion	Six of ten gates failed; budget coverage and host instability explained much of the apparent benefit.
S5/S6	Survives, scoped	Safe spectral cuts passed the declared moving-frame sweep; task benefit remains topology-dependent.
S7	Correction	Curvature, global holonomy, and arbitrary sheaf H^1 are not one equivalence.
R6	Survives, bounded	Exact representation cost and finite simulation checks; no hardware or advantage claim.
S8	Bounded pass	Fano 7/21/28 identifiability survives; unique comparative rank interpretation does not.
S9	Killed parameter point	Instrument corrected; strict bundle isolation still failed.

THE PATTERN WORTH EXPORTING

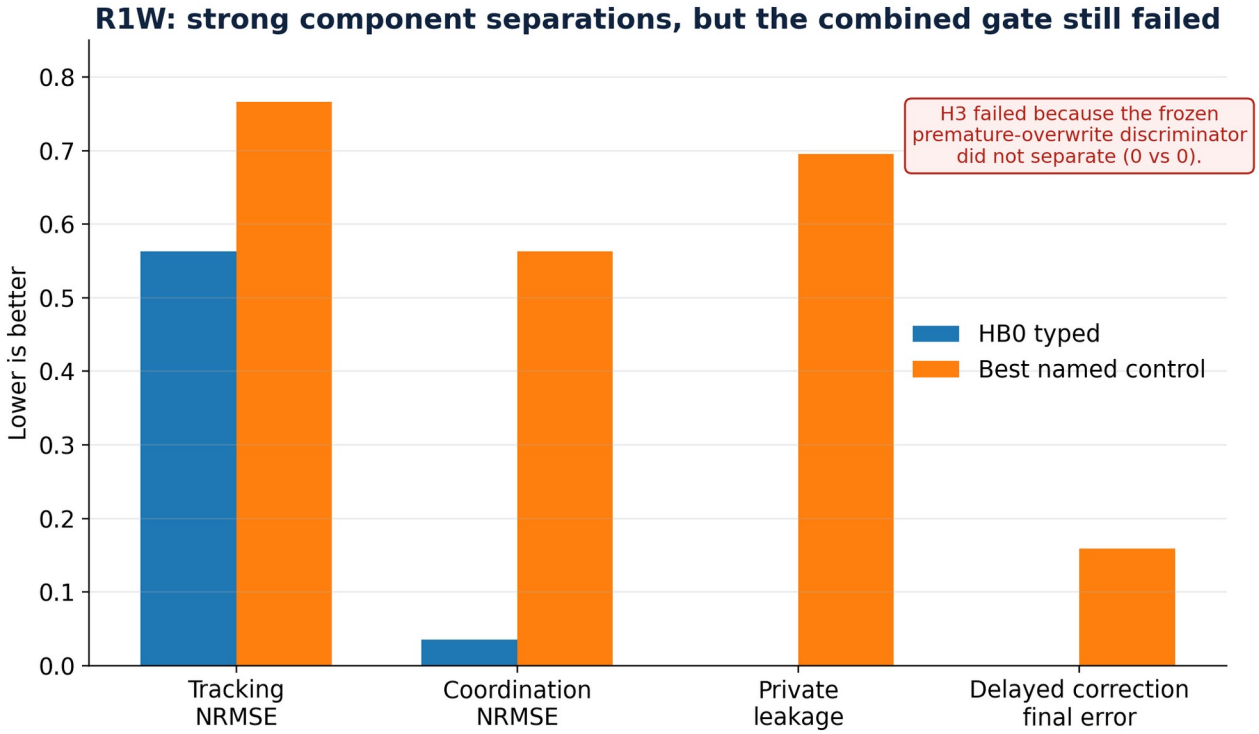
The project is most mature when a failed result makes the next instrument harder to fool. That is a general research-engineering capability, independent of whether any specific “brain” hypothesis eventually succeeds.

Repository source trail: docs/S4_G100_RESULTS.md; README.md research ledger; docs/ARCHITECTURE.md; docs/MANIFOLD_REFERENCE.md

CHAPTER 18

R1W bounded competence

Two strong component results, one failed preregistered discriminator, and no combined promotion.



Selected R1W component values. Lower is better. The chart is illustrative; the verdict is governed by the frozen multi-part gate, not by visual effect size.

R1W is the repository's clearest attempt to connect HB-0 to task performance. It runs synthetic latent-state tracking under chart drift, mutual-recognition coordination with private information, and delayed correction. The controls include a flat pooled model, a shuffled typed model, and a one-clock local model. The evaluation spans N = 16, 32, and 64 with twenty deterministic seeds per size.

The tracking component passed. Typed per-Worldlet transport calibration reduced reconstruction error against both the pooled chart and the correspondence-shuffled typed null, with matched matrix-vector multiply counts. The coordination component also passed: the typed protocol reconstructed shared state while its declared non-absorption mask prevented private leakage.

The delayed-correction component did not pass its full gate. HB-0 achieved zero final error while the one-clock control retained error, but the frozen gate also required the control to exhibit at least 0.20 more premature slow overwrites. Both observed rates were zero under the preregistered definition. No threshold was changed after the fact.

Component	Verdict	Strongest authorized statement	Why it stops there
H1 tracking	PASS	Typed local transport improves reconstruction on the constructed chart-drift family at matched multiply count.	The synthetic task may favor the architecture's own typing assumptions.
H2 coordination	PASS	The declared shared/private protocol coordinates without transmitting the private coordinate.	The non-absorption mask is designed, not emergent social cognition.
H3 delayed correction	FAIL	Final error improved, but the temporal mechanism was not established.	The frozen premature-overwrite discriminator did not separate.
Combined gate	FAIL	No promoted "bounded polycosmic competence" claim.	All preregistered components were required.

WHY THE COMBINED FAILURE MATTERS

A weaker reporting culture could advertise two wins out of three and move on. R1W keeps the combined claim unpromoted because the mechanism it wanted to demonstrate was part of the question, not an optional explanatory story.

What R1W changes

The older HB-0 design document correctly said that no preregistered held-out competence effect existed at that time. R1W updates that history without turning the architecture into an AGI result. There are now bounded measured component effects on constructed tasks. There is still no external cognitive benchmark, no independent witness, no physical quantum substrate, and no combined promotion.

The most useful next move is therefore discriminating rather than decorative: transfer the strongest components to task families not built around the architecture's own coordinate assumptions, keep compute and capacity matched, and preserve the safety and identity gates that make the comparison meaningful.

Repository source trail: docs/HB0_R1W_COMPETENCE_RESULT.md; experiments/run_hb0_r1w_competence.py; tests/test_hb0_r1w_competence.py

CHAPTER 19

R1X to R1Z: the live frontier and the criterion problem

The frontier moved from a spectral measure, to its hyperellipsoid body, to a size-dependent quasi-free-state question - and then found its own verdict rule in the way.

CHAPTER ABSTRACT

This chapter follows three correction-preserving rounds. R1X introduced a Lebesgue reading of restricted covariance spectra but left the step budget free. R1Y named the Williamson hyperellipsoid and recovered orientation, while exposing that most axes had already degenerated. R1Z moved to growth with register size and the Araki/Powers-Størmer category, but its BOUNDED rule was unreachable for genuinely bounded-looking sequences and its frozen zero-count observable required full degeneracy. The branch's strongest frontier result is therefore not an asymptotic theorem; it is a concrete demonstration that preregistration must validate its decision procedure before it can govern evidence.

R1X: a better carrier, an under-specified decision

R1X represented the symplectic values of a restricted fermionic Gaussian covariance as an empirical measure. Entanglement entropy became an integral over that measure, purity mass summarized collapse toward an atom, and a Wasserstein reconstruction defect measured failure to rebuild the whole from its parts. The frame-invariance control was exact linear algebra, as it should have been.

The discriminating prediction failed. More importantly, the preregistration froze two thresholds but described the step budget only as "fixed" without assigning a value. Results at 300 and 900 steps pointed in different directions. A free parameter inside a frozen protocol had become a threshold in disguise. The round preserved both budgets and recorded the defect rather than choosing the favorable one.

R1Y: name the body, not only its axis measure

Williamson's theorem identifies the restricted covariance with a symplectic hyperellipsoid whose symplectic values are its semi-axis lengths. R1X measured the lengths and forgot orientation. R1Y correctly asked whether the frame retains information after the lengths degenerate, and whether an intrinsic GF(16)-derived tessellation replaces the arbitrary left-half mode cut.

The preregistration also corrected R1X's causal explanation. Across the R1X sweep, 81 of 84 semi-axes were exactly 0 or 1. A measure supported on {0,1} has only one degree of freedom, so the spectral measure could not out-inform the scalar in six of seven cells. The budget defect remained real, but it was not the mechanism of failure.

R1Z: move the limit question onto the n-axis

R1Z rejected an initially tempting but category-wrong analogy to Feldman-Hájek Gaussian measures. Fermionic Gaussian states are quasi-free states of the CAR algebra. For the gauge-invariant case, the Powers-Størmer/Araki criterion against the Fock vacuum reduces to finiteness of expected occupation. The repository then measured finite-size growth as evidence that could be consistent with one side or the other, while explicitly forbidding a finite-n claim about the limiting dichotomy.

Herald family	Median zero counts over n = 4..32	Frozen slope	Frozen verdict
ALL_PAIRS	2, 2, 2, 3, 6, 6, 8	0.482	DIVERGENT
NEIGHBOUR	0, 0, 1, 1, 0, 1, 1	0.051	INDETERMINATE
RANDOM_MATCHING	2, 2, 2, 5, 4, 6, 8	0.436	DIVERGENT; unpredicted
BLOCKED_M3	0, 0, 2, 1, 0, 1, 0	-0.032	INDETERMINATE
BLOCKED_FROBENIUS	0, 0, 0, 1, 0, 0, 0	-0.020	INDETERMINATE

All three gates passed, global purity defect remained below 6.8e-14, and P3 held for ALL_PAIRS. RANDOM_MATCHING, deliberately left unpredicted, also showed a positive growth slope despite being as sparse as NEIGHBOUR. On this finite testbed, locality rather than pair density tracked the separation. That is a reading of five constructed families, not a general law.

The criterion was the most important result

The frozen BOUNDED rule required both slope < 0.05 and largest <= 2 * max(smallest, 1e-12). For a genuinely bounded family that is zero on small registers and one on some larger register, the second clause becomes largest <= 2e-12 and is impossible. The rule could not certify the sequences it was designed to recognize. Its slope threshold was also at roughly the one-count resolution of the fitted grid, so only an exactly constant sequence could pass robustly.

A second defect concerned the observable. The Araki criterion is naturally expressed through expected occupation, which the implementation recorded on every row. The preregistered zero count equals entropy only when every semi-axis is fully degenerate at 0 or 1. Thirty-two of the 105 rows did not satisfy that precondition. Occupation separated the families more cleanly, but it was reported post hoc and moved no verdict.

PEER-REVIEW LESSON FROM R1Z

Preregistration is not enough. Before a frozen criterion is allowed to decide a scientific result, the criterion itself should be exercised against synthetic examples of every verdict it claims to produce, including boundary cases, zeros, resolution limits, and known refutations. R1Z shows that an immutable bad rule can be as misleading as an adjustable good-looking one.

What survives and what does not

Status	Surviving statement
P / exact within the encoded objects	The bridge identity on constructed fully degenerate covariances; finite matrix/covariance calculations; purity checks.
M	The five-family growth table over seven register sizes, three seeds, and 400 steps; the reported local environment and raw artifact.
H	Any claim about bounded versus divergent occupation as n approaches infinity; any quasi-equivalence or disjointness conclusion.
R / corrected	The R1X causal attribution to budget as the mechanism; the R1Z BOUNDED decision rule as a valid classifier.
Not independently witnessed	Producer, implementation, tests, and interpretation remain on one development line; no cross-platform job is claimed.

The bounded next round should preregister occupation rather than zero count, unit-test the decision rule on synthetic bounded, divergent, indeterminate, zero-leading, and resolution-edge sequences, and stop before any richer architecture is added. The criterion preflight is the experiment's first gate, not an after-action correction.

Repository source trail: manifests/qsigma_r1x_spectral_measure_finding.json; manifests/qsigma_r1y_hyperellipsoid_tessellation_preregistration.json; manifests/qsigma_r1z_araki_dichotomy_preregistration.json; manifests/qsigma_r1z_araki_dichotomy_finding.json; src/aionq/araki.py; tests/test_araki.py; commit fbf7b9b

PART VI

Reproducibility and use

What is formal, what is merely bound, and how a new reader can run or extend the work

observe | verify | leave the remainder visible

CHAPTER 20

Formalization, envelopes, and witnesses

A successful build, a valid envelope, and an independent result are three different claims.

The Lean surface is mixed

The formal directory contains repository-local Lean files plus a vendored archive. The local status note is explicit: OhniumObservatory.lean has a narrow direct acceptance record under a local Lean toolchain. RecognitionChamber.lean has not compiled successfully in the recorded environment because Mathlib is unavailable, and it contains a sorry in the finite-depletion target. The vendored leanmath archive is not a completed, buildable proof corpus.

The accepted Ohnium file proves only a small cadence and cardinality surface under explicit hypotheses. It does not formalize the Hopf map, quantum Fisher information, adiabatic dynamics, cognition, or a physical observatory. Numerical P-tagged claims elsewhere must not be retroactively described as Lean-verified.

Surface	Recorded status	Authorized reading
formal/OhniumObservatory.lean	Direct local acceptance, narrow dependency surface	A small exact formal artifact with limited scope.
formal/RecognitionChamber.lean	Dependency-blocked; contains one sorry	Open formal target, not a closed proof.
formal/leanmath/	Vendored; missing modules and parse/build issues	Provenance context, not evidence for this repository.
Repository as a whole	No fresh full formal build in this guide	Do not infer global theorem status from selected files.

Evidence envelopes

An evidence envelope binds an artifact to named code, inputs, environment, result, and witness state. Verification can detect changed files, digest mismatches, missing effective inputs, or a claim/tag mismatch. This is strong provenance engineering.

Envelope validity is not independent replication. A producer can correctly verify its own output. The repository keeps independence as a separate axis and uses historical immutable blobs only under explicit fallback rules. The guide did not run the envelope verifier against HEAD, so current counts from older reviews are not repeated as fresh facts.

Witness weave

The witness scheduler models parties as vertices and direct verification acts as edges. Evidence rank is the strongest direct act for a claim, not the sum of a social path. Attestation rings cannot inflate reviewed or independently recomputed status. Pending conversation handles remain pending until an explicit response is ingested.

The append-only journal splits crash boundaries and refuses automatic retries. This is unusually honest agent infrastructure: a conversation URL is a transport handle, not evidence; asynchronous execution is not a promise that this session will later collect a verdict.

THREE QUESTIONS TO KEEP SEPARATE

1) Did the code execute? 2) Does the encoded statement support the intended external claim? 3) Did an independent party recompute it? HypercomplexBrain has separate mechanisms for all three because collapsing them is a major source of false confidence.

Repository source trail: formal/README.md; formal/LEANMATH_PROVENANCE.md; src/aionq/evidence.py; docs/WITNESS_WEAVE.md; docs/HANDOFF.md

CHAPTER 21

Run, navigate, and extend

A quiet path for readers, an audit path for researchers, and a strict path for new rounds.

Local setup

Repository entrance commands

```
python -m venv .venv

# macOS / Linux
.venv/bin/python -m pip install -e ".[dev]"
.venv/bin/python -m pytest -q
.venv/bin/python experiments/run_audit.py

# Windows PowerShell
.\.venv\Scripts\python.exe -m pip install -e ".[dev]"
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe experiments/run_audit.py
```

The two platform blocks are alternatives. Read a round document before executing its producer. Some outputs are append-once or versioned and will correctly refuse to overwrite existing evidence. A refusal is often a governance success, not a runtime failure.

The repository-wide pytest census changes as research branches move, so the README intentionally avoids freezing a count. Treat a reported historical count as belonging to its commit and environment.

Verify evidence separately

Provenance verification

```
PYTHONPATH=src python -m aionq.verify_envelopes artifacts
```

This checks the evidence membrane. It does not rerun every scientific computation, and a first-party envelope remains first-party. For a deterministic result, independent recomputation should compare digests under a declared compatible environment. For timing or environment-sensitive results, methodological review may be more meaningful than byte equality.

A disciplined way to add a research round

1. Name one object and one bounded question. Do not begin from a desired conclusion.
2. Define the measurable operator, its units, frame, free theory, and numerical domain.
3. Freeze predictions, thresholds, budgets, seeds, controls, and kill conditions in a manifest before implementation or execution.

4. Include at least one control capable of producing an embarrassing answer. Match compute and capacity where the comparison demands it.
5. Implement raw measurements and typed remainders before convenience scores.
6. Make the runner fail closed on invalid numerics, incomplete controls, or attempted overwrite.
7. Bind artifacts to provenance and report witness independence separately.
8. Promote only the exact scoped statement. Preserve failures and supersede interpretations instead of rewriting history.

THE MOST USEFUL CONTRIBUTOR BEHAVIOR

Do not add another decorative module until the current architecture has a discriminating benchmark or theorem obligation. In this repository, depth comes from tighter bridges and better controls, not from the number of exotic nouns.

Suggested reading and execution routes

Goal	Read first	Then run
Understand the architecture	README -> HB0_POLYCOSMIC_KERNEL -> HYPERBOLIC_BASE	tests/test_hb0.py and focused HB0 tests
Audit evidence governance	SOURCES_DIGEST -> WITNESS_WEAVE -> formal/README	verify_envelopes and governance tests
Study quantum cost	QUANTUM_SUBSTRATE_COST -> OPERATIONAL_INFORMATION	run_hyperbolic_qubit_cost.py and focused tests
Study bounded competence	HB0_R1W_COMPETENCE_RESULT + frozen manifest	run_hb0_r1w_competence.py and its tests
Continue the frontier	R1X finding -> R1Y preregistration -> R1Z preregistration and finding	Preflight the next decision criterion on synthetic verdict cases before any producer runs

Repository source trail: README.md (Run and Explore); pyproject.toml; docs/WITNESS_WEAVE.md; manifests conventions

PART VII

Assessment

What is strongest now, what remains open, and what would turn the architecture into a deeper scientific result

observe | verify | leave the remainder visible

CHAPTER 22

The strongest surviving claims

The most defensible achievements are structural, methodological, and narrowly empirical.

1. Evidence-governed research architecture

The repository implements a rare end-to-end claim discipline: explicit evidence types, frozen manifests, controls, append-only artifacts, provenance envelopes, witness state, and supersession. It has repeatedly used that machinery to lower its own claims. This is stronger evidence of the governance system than any README declaration could provide.

2. Anti-flattening structures with executable refusal

Worldlets preserve local geometry; Tendrils report losses; quality channels reject weighted sums; vector Hamiltonians prevent cross-channel cancellation; higher cells retain obstructions that edges can miss; all Fin(n) sectors are reconstructable; missing evidence remains a typed remainder. These are real implementation properties, even though their cognitive value remains open.

3. Scoped mathematical and numerical results

The branch contains bounded results that survive their declared controls: safe-cut moving-frame invariance in the reported sweep, exact representation identities for the unitary price of depth, a well-conditioned Fano 7/21/28 local coordinate system, a repaired chiral spin transport, and component-level R1W effects on constructed tasks. Each result has an interpretation ceiling, which is part of the result rather than an afterthought.

4. Preserved negative knowledge

The repo knows several things that do not work: scalar quantumness, a basis-sovereign reading of S3 localization, the S4 chamber's combined promise, an overbroad holonomy-curvature-H^1 chain, the frozen S9 parameter point, the R1W delayed-correction mechanism, and the late-time R1X separation. This negative map reduces the space of future self-deception.

Evidence class	Strong current example	What it does not imply
P / exact within model	Representation identities, algebraic ranks, typed refusal properties	External physical or cognitive truth without a bridge
A	Shared H^4 base and weaker theorem surrogates	Empirical selection or uniqueness
M	Finite sweeps, R1W components, S8/S9 numerics	Asymptotic law, general benchmark superiority, independent replication

Evidence class	Strong current example	What it does not imply
H	R1Z asymptotic quasi-free-state growth interpretation; HB-0 competence and recognition hypotheses	A theorem about the limit, a cognitive result, or independent replication
S	Worldlet / identity / recognition architecture	Sentience, AGI, or biological ontology
R	Scalar and basis-dependent interpretations	A live claim that may be cited as current

OVERALL ASSESSMENT

The repository is already a serious research artifact because it exposes the difference between expressive architecture and warranted conclusion. It is not yet evidence that the full architecture thinks better than simpler systems. That remaining gap is clearly named rather than obscured.

Repository source trail: README.md current verdicts; docs/HB0_POLYCOSMIC_KERNEL.md; docs/HB0_R1W_COMPETENCE_RESULT.md; R1Z preregistration, finding, artifact, and tests

CHAPTER 23

The smallest honest road to a brain

The next breakthroughs should connect representation to competence without weakening the evidence membrane.

The repository has already shown that it can generate rich structure and test local invariants. The open scientific question is whether those structures produce a repeatable advantage on tasks not designed to mirror them. That bridge should be narrower and more demanding than “build a larger architecture.”

A credible programme would preserve the current philosophy while changing the center of gravity from internal structure to external discrimination. The architecture should face environments with hidden state, shifting charts, private information, delayed consequences, and irreversible correction - but the task generator, controls, and scoring policy should be fixed before the full HB-0 model is allowed to see them.

Five next obligations

1. Independent task design. Use at least one benchmark family authored outside the architecture’s own vocabulary, so typed transport is not rewarded merely because the generator was built around typed charts.
2. Capacity- and compute-matched controls. Compare full HB-0 against flat pooled, dynamic graph, real-carrier, fixed-hypercomplex, one-clock, and ablated variants under equal budgets.
3. Mechanism-specific gates. When a multi-clock or recognition mechanism is claimed, require a discriminator that only that mechanism should satisfy, not just a lower final error.
4. Independent reproduction. Prioritize deterministic rounds whose artifacts can be recomputed by another environment and whose digest disagreement would be meaningful.
5. Bridge the formal and empirical surfaces. Formalize the exact invariance or refusal property that the experiment depends on, while keeping the task-performance claim measured rather than laundering it through theorem language.

What not to do next

- Do not promote hypercomplex dimension itself as intelligence or computational escape.
- Do not replace failed componentwise gates with a more forgiving scalar.
- Do not infer a global phase, capacity, or asymptotic law from one finite parameter point.
- Do not treat a valid envelope, a passing test, or an AI review as independent corroboration.
- Do not smooth over the friction where an object degenerates, a frame matters, or a control wins; those are the places where understanding grows.

THE HUMAN MEANING OF THE PROJECT

HypercomplexBrain is ultimately an attempt to build systems that can move among partial worlds without pretending that translation is lossless, that one viewpoint is sovereign, or that uncertainty is zero. Its scientific worth will depend on whether that ethical and geometric intuition can be converted into measurable competence under controls that are strong enough to say no.

Closing perspective

In an age of abundant machine-generated proofs, code, and architectures, the scarce achievement is not another plausible construction. It is a construction whose meaning remains legible after the controls run. This repository is at its best when it makes the hard parts visible: the frame that moved, the budget that was free, the axis that had already collapsed, the proof file that did not compile, the parameter point that stayed killed, and the combined gate that refused two attractive partial wins.

That is why the correction ledger is not an appendix. It is the architecture's memory of contact with reality. The next human-readable edition should become shorter in explanation not because the system is simpler, but because the underlying operators, controls, and external results have become clearer.

Repository source trail: Synthesis of the inspected branch at commit fbf7b9ba8974364ee212a57e347ab0d0de37e454

APPENDICES

Reference atlas

Terms, modules, commands, evidence status, and the exact repository paths used for this edition

observe | verify | leave the remainder visible

Appendix A - Glossary for human readers

Term	Plain-language meaning
AION-Omega^Q / QSigma	The umbrella architecture and its sequence of quantum-evidence revisions.
Worldlet	A typed local context that keeps carrier, geometry, algebra/observability, clocks, observers, boundary, energy/phase, quality channels, provenance, and typed remainders together. The kernel separately assigns its shared-base point. It is not a neuron, agent, universe, or sentient subject.
Tendril	A declared correspondence from one Worldlet to another. It reports a forward map, an optional return map, what is preserved/transformed/lost, cost, raw round-trip defect, and [R]_transport. Low defect is local reconstruction, not semantic equivalence.
CosmosCage	An acyclic nested scope container. The code refuses cycles and duplicate cage IDs; it does not yet prove a topological cover or unique Worldlet membership.
GerbeCell	A supplied triple of square transition matrices and a comparison matrix whose Frobenius closure defect is measured. It is a coherence witness, not automatically a gerbe, cocycle, or cohomology class.
Polyholonomic identity	An identity ecology of local regime sections, local holonomies, comparison maps, higher-coherence witnesses, viability gates, and [R]_id. Its summaries are engineering heuristics, not evidence of selfhood.
Free theory	The class of structure treated as cheap or available for a comparison; quantumness is relative to it.
Frame	A labeling, basis, gauge, chart, or coordinate system through which an object is measured.
Orbit-collapsed	Invariant under the declared frame action at the tested level.
Orbit-resolving	Changes with the frame and must be interpreted as a coordinate reading.

Term	Plain-language meaning
Quantum Necessity Atlas	Vector-valued, interval-aware, free-theory-relative measurement surface.
Promotion cone	A componentwise acceptable region; no weighted sum may hide a failed dimension.
Typed remainder [R]	An explicit unresolved loss or limitation whose kind is part of its type.
Bordón	Reservoir model inspired by the continuous drone of a Galician gaita.
Holoportation	Reconstruction from a boundary or transported representation while preserving nullity and error.
Projection violence	Information or distinction destroyed by a representation or observation map.
Hypercomplex carrier	Complex, quaternionic, octonionic, or higher algebra used as observability/operator structure.
Higher cell	An n-ary relation that may contain information invisible to every pairwise edge.
Fin(n) sectors	The complete finite Fourier family retained rather than reduced to a mean or dominant component.
Evidence envelope	Provenance binding between claim, code, inputs, environment, artifact, and witness state.
Witness weave	Append-only scheduler and nerve for external review or recomputation without evidence inflation.
Supersession	A correction relation that preserves the earlier claim and points to the later governing interpretation.
Fail-closed	Missing, invalid, or uncontrolled evidence blocks promotion rather than defaulting to success.
ACAF attack	An adversarial control or fuzzing surface designed to find representation, parameter, or governance defects.
QSigma-R1Y	The current preregistered frontier on hyperellipsoid orientation and intrinsic tessellation survival.

Appendix B - Module atlas

Family	Representative modules	Purpose
Evidence and governance	governance.py, evidence.py, verify_envelopes.py, witness*.py	Claim types, promotion rules, provenance, witness states, refusal paths.
Atlas and comparison	atlas.py, advantage.py, quantities.py, stack.py, residency.py	Typed coordinates, intervals, free theories, transport, simulation-cost bounds.
Frames and geometry	hyperbolic.py, movingframe.py, frame_orbit.py, connection_frame.py	H^n base, Euclidean ablation, group actions, safe comparison.

Family	Representative modules	Purpose
HB-0 core	hb0/core.py, kernel.py, manifest.py, transport.py	Worldlets, Tendrils, cages, construction and review.
HB-0 identity and decision	hb0/identity.py, mind.py, decision_field.py, wall.py	Identity ecology, quality field, higher cells, n-ary walls.
Quantum surfaces	hb0/quantum_base.py, hb0/chiral_spin.py, coherentize.py, fqhe.py	Typed point/state shadows, chiral transport, coherentization, finite Chern probe.
Algebraic surfaces	cayley_dickson.py, fano_algebra.py, fano_decoder.py, braid.py, fusion.py	Associator regimes, algebra-valued channels, braid and fusion checks.
Memory and resource	process.py, bordon.py, budget.py, equal_resource_reservoir.py, capacity_ceiling.py	Temporal memory, reservoir flow, matched resources, ceilings.
Information and reconstruction	visibility.py, selfknowledge.py, closure.py, contraction.py	Local visibility, self-knowledge bounds, closure, classical strategy costs.
Experimental organs	recognition/, four_cell.py, carrier_telemetry.py, common_receiver_reservoir.py	Bounded chamber, calibrated roles, telemetry, receiver/reservoir studies.

Appendix C - Experiment command map

Surface	Command	Purpose
Core audit	python experiments/run_audit.py	Atlas, transport, fusion, identity, budgets, correction contracts
Falsification sweep	python experiments/acaf_sweep.py	Representation and cancellation attacks over generated cores
Visibility	python experiments/run_visibility.py	Local-probe coupling field and controls
Replication	python experiments/run_replication.py	Depth, intervals, and distributions
S1 Sentinel	python experiments/run_sentinel.py	Sparse-gap novelty claim and kill condition
S2 Stokes	python experiments/run_stokes.py	Braid action and basis mutation
S3 Sheaf	python experiments/run_sheaf.py	Obstruction localization and matched controls
S4 Recognition	python experiments/run_s4_g100.py	Ten preregistered chamber gates
S4 Basis attack	python experiments/run_s4_basis_attack.py	Frame dependence of S3 instrument
S5 Moving frame	python experiments/run_moving_frame.py	Orbit collapse versus orbit resolution
S6 ACAF frame sweep	python experiments/run_acaf_frame_sweep.py	Degenerate-cut adversarial search
S7 Adjudication	python experiments/run_audit_adjudication.py	Recompute external corrections
R6 substrate cost	python experiments/run_hyperbolic_qubit_cost.py	Compact/noncompact representation cost

Surface	Command	Purpose
S8 Fano decoder	python experiments/run_fano_decoder.py	Algebra-valued local identifiability
R1W competence	python experiments/run_hb0_r1w_competence.py	Tracking, coordination, delayed correction
Evidence verifier	python -m aionq.verify_envelopes artifacts	Artifact and envelope provenance checks

S9 AND APPEND-ONCE ROUNDS

Use the exact frozen commands in the round document and manifest. Some producers are intentionally versioned or non-overwriting. Do not generalize the command map into permission to rerun every artifact in place.

Appendix D - Current evidence matrix

Object	Status	Strongest support	Active ceiling
Quantum Necessity Atlas	ACTIVE / structural	Vector, interval, mask, transport, and promotion machinery	No causal or cognitive advantage by itself
Shared H^4 base	A	Repository-wide coordinate contract with Euclidean ablation	Not empirically selected as optimal
Moving-frame safe cuts	M/P mixture, scoped	Reported 950/950 adversarial invariance in frozen sweep	Accuracy remains topology-dependent
R6 substrate cost	P + M, bounded	Exact depth/cost identities and finite simulations	No hardware or quantum advantage
S8 Fano decoder	M, bounded pass	7/21/28 identifiability in linearized so(8)	Comparative rank advantage corrected
S9 FQH point	R for the point	Frame repaired; strict isolation still failed	No C/2 response or phase claim
HB-0 architecture	S/H + executable structure	Worldlets, transports, qualities, clocks, higher cells	No AGI, sentience, or global competence result
R1W components	M, self-attested	Tracking and coordination effects on constructed tasks	Combined promotion failed; no external benchmark
Formal surface	Mixed	Narrow Ohnium acceptance; other surfaces incomplete	No globally verified theorem corpus
R1Z growth experiment	M + H; criterion corrected	Five-family finite-size measurements; gates and purity checks; defective bounded rule preserved	No asymptotic quasi-equivalence/disjointness, area/volume-law theorem, or independent witness

Appendix E - Repository source map for this edition

Topic	Primary repository paths
Entrance and current posture	README.md
Main architecture and scalar corrections	docs/ARCHITECTURE.md
Evidence governance and imported boundaries	docs/SOURCES_DIGEST.md
HB-0 core	docs/HB0_POLYCOSMIC_KERNEL.md
Universal base	docs/HB0_HYPERBOLIC_BASE.md
N-ary decision field	docs/HB0_HYPERCOMPLEX_DECISION_FIELD.md
Chiral quantum fiber	docs/HB0_CHIRAL_SPIN_CONNECTION.md
Bounded competence	docs/HB0_R1W_COMPETENCE_RESULT.md
Substrate price	docs/QUANTUM_SUBSTRATE_COST.md
Operational information	docs/OPERATIONAL_INFORMATION.md
S4 result and S3 correction	docs/S4_G100_RESULTS.md
Fano decoder	docs/S8_FANO_DECODER.md
Finite Hofstadter probe	docs/S9_FQHE_PROBE.md
Witness system	docs/WITNESS_WEAVE.md; docs/HANDOFF.md
Formal status	formal/README.md; formal/LEANMATH_PROVENANCE.md
R1X finding	manifests/qsigma_r1x_spectral_measure_finding.json
R1Y frontier	manifests/ qsigma_r1y_hyperellipsoid_tessellation_preregistration.json
Package surface	src/aionq/__init__.py; pyproject.toml
R1Z finite-size growth and criterion correction	manifests/qsigma_r1z_araki_dichotomy_preregistration.json; manifests/qsigma_r1z_araki_dichotomy_finding.json; src/aionq/araki.py; tests/test_araki.py

CITATION NOTE

This local guide synthesizes repository material rather than reproducing it verbatim. The authoritative statements remain the committed manifests, result artifacts, tests, formal files, and supersession links at the inspected snapshot.

Appendix F - Conceptual lineage and non-equivalences

Custom vocabulary becomes easier to evaluate when placed beside established ideas without declaring identity. The table below gives the nearest reading aid and the difference that must remain visible.

Custom concept	Nearby established idea	Crucial non-equivalence
Worldlet	Local chart, typed module, context object, partial model	No theorem shows that the Worldlet collection is a sheaf, a cognitive module decomposition, or an optimal factorization.
Tendril	Functor, profunctor, encoder/decoder, lens	The executable object is a declared pair of numerical maps plus receipt metadata; categorical laws are not automatic.
CosmosCage	Nested scope, tree decomposition, cover	Acyclic nesting is implemented; topological cover axioms and unique membership are not.
GerbeCell	Cocycle/coherence equation on triple overlaps	A finite supplied matrix defect is measured; site, coefficient object, gauge, and cohomology remain open.
Polyholonomic identity	Bundle sections and holonomy; multi-context state	The analogy organizes software state; it does not establish a geometric theory of personal identity.
Recognition without absorption	Multi-agent belief modelling; information-preserving alignment	The formula is a design score pending validation against simpler coordination metrics.
Holoportation	Boundary reconstruction / inverse problem	Moore-Penrose reconstruction reports nullity; it is not physical teleportation or guaranteed semantic recovery.
Mind-quality field	Multi-objective control / constrained optimization	The names are custom; the current scientific content lies in the operators, obligations, and refusal to scalarize.

READER SAFEGUARD

Analogy is a generator of candidate mathematics, not a bridge theorem. Whenever a custom term resembles a standard object, ask which axioms, invariants, and equivalences have actually been implemented or proved.

Colophon

Prepared as a local publisher-review edition from claude/aion-brain-architecture-27f8ji at fbf7b9ba8974364ee212a57e347ab0d0de37e454. The GitHub repository was not modified.

Document design: reader-first technical handbook with preserved evidence boundaries, failure modes, and repository source trails. Diagrams are explanatory reconstructions based on the committed architecture; they are not experimental artifacts.

Primary maxim: observe, verify, and leave the remainder visible.